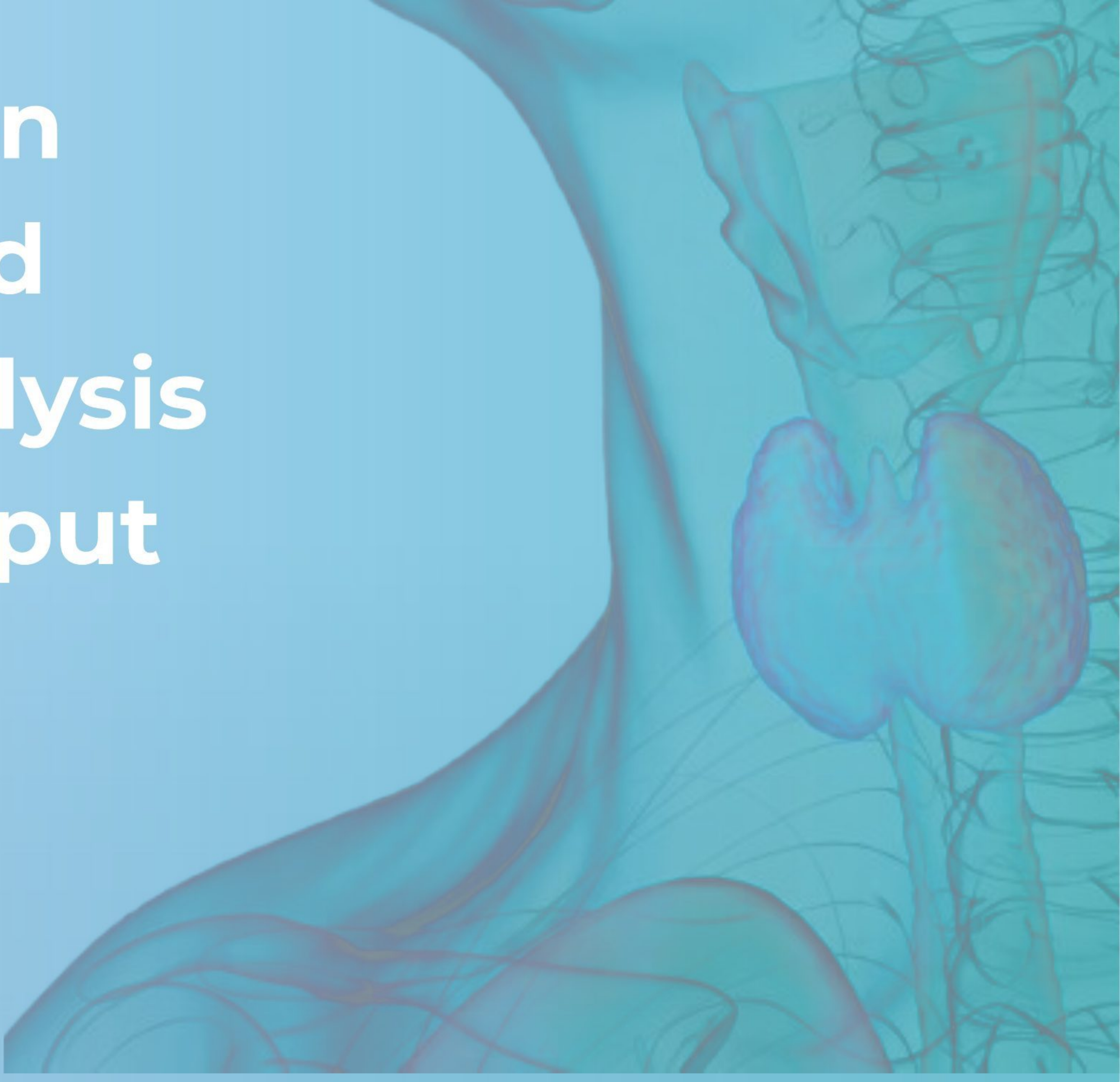




Patient-centered Optimization Design of I-131 Treatment Services for Thyroid Cancer

XIAO Yucheng XU Danni HE Peizhen

- 
- 01 Introduction
 - 02 Background
 - 03 Design analysis
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Current situation



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Why thyroid cancer ?

According to the 2025 projections of the American Cancer Society (ACS):

Research Insights

44020

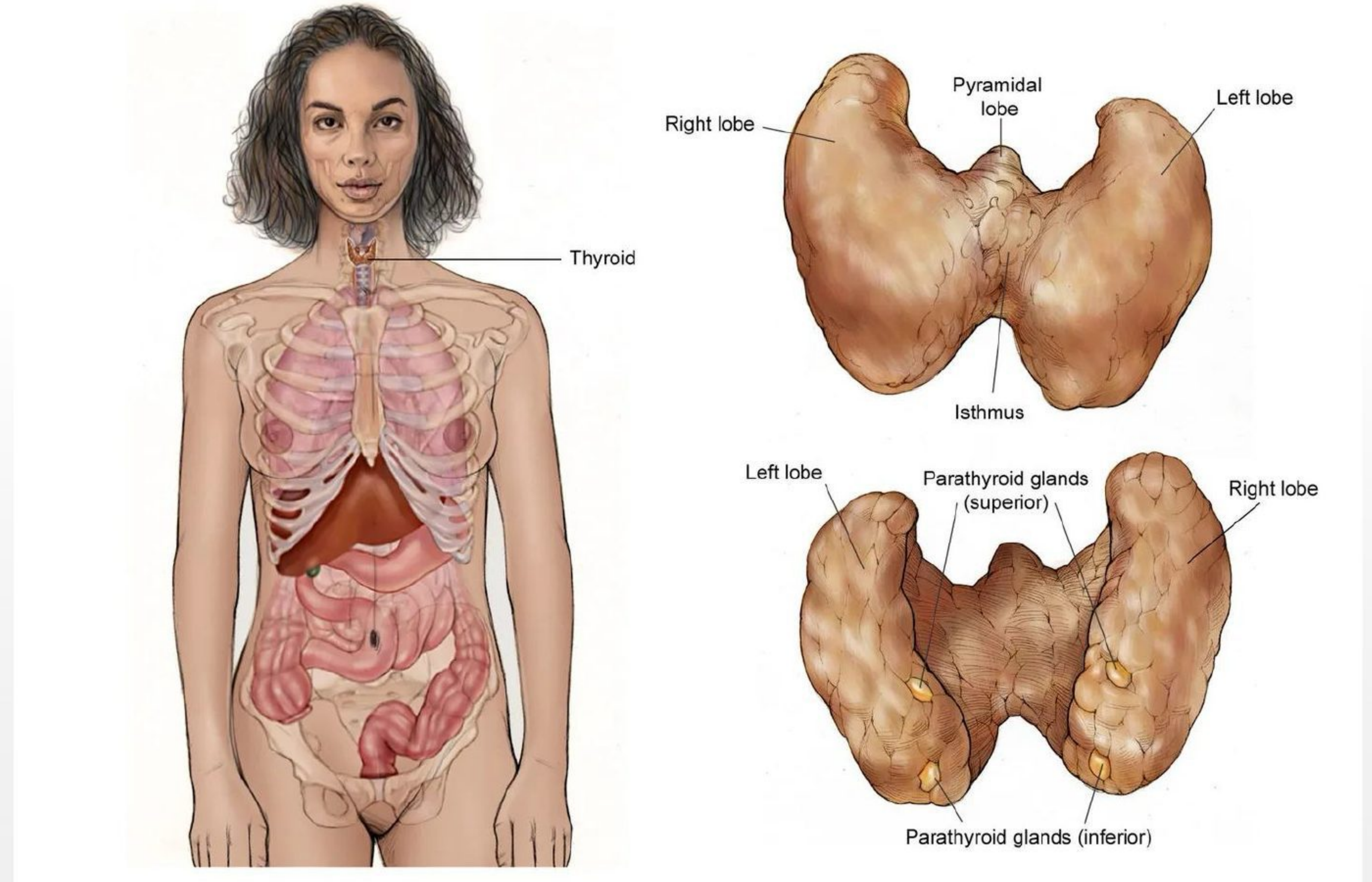
The number of new cases is approximately **44,020**.

12th

It ranks around the **12th** position among all cancers (more prominent among women).

Fourth

If the current trend continues, it may become **the fourth most common cancer** in the United States by **2030** (predicted by ACS & NCI).



Why thyroid cancer ?

According to the "2024 China Malignant Tumor Epidemiological Report" by the National Cancer Center:

Research Insights

44610

In 2024, there were approximately **466,100** new cases of thyroid cancer (466,100 cases) across the country.

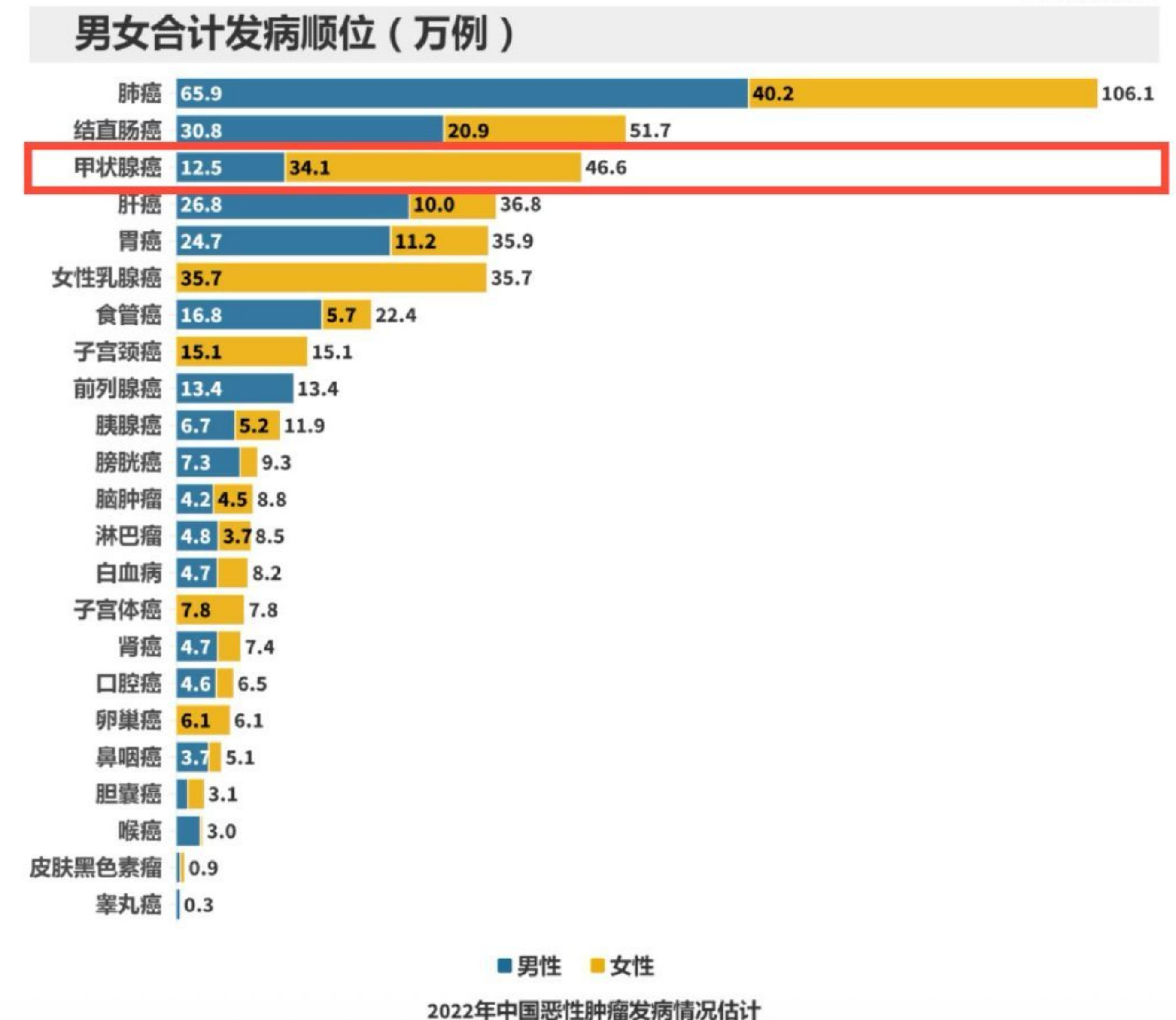
Third

It ranks **third** among all malignant tumors, following lung cancer and breast cancer.

The country with the heaviest burden globally.

It accounts for nearly **80%** (approximately 79.5%) of new thyroid cancer cases worldwide.

The leading cause of new cancers
among individuals aged 15-29.



Three major trends in thyroid cancer

According to above information



The incidence rate
Rising rapidly



The mortality rate
**Continuously
decreasing**



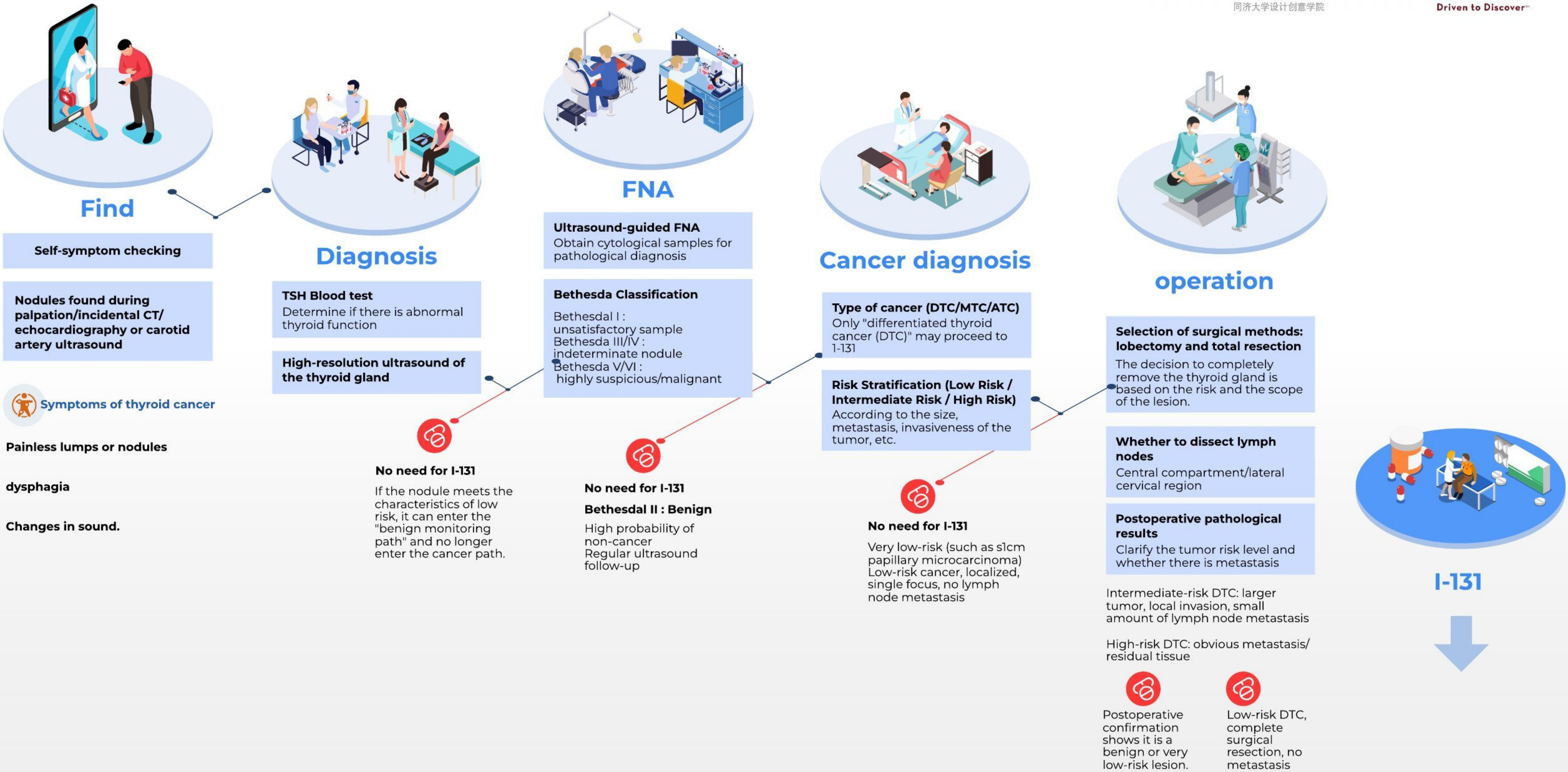
Trend of Y population
Obvious

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Thyroid cancer treatment process



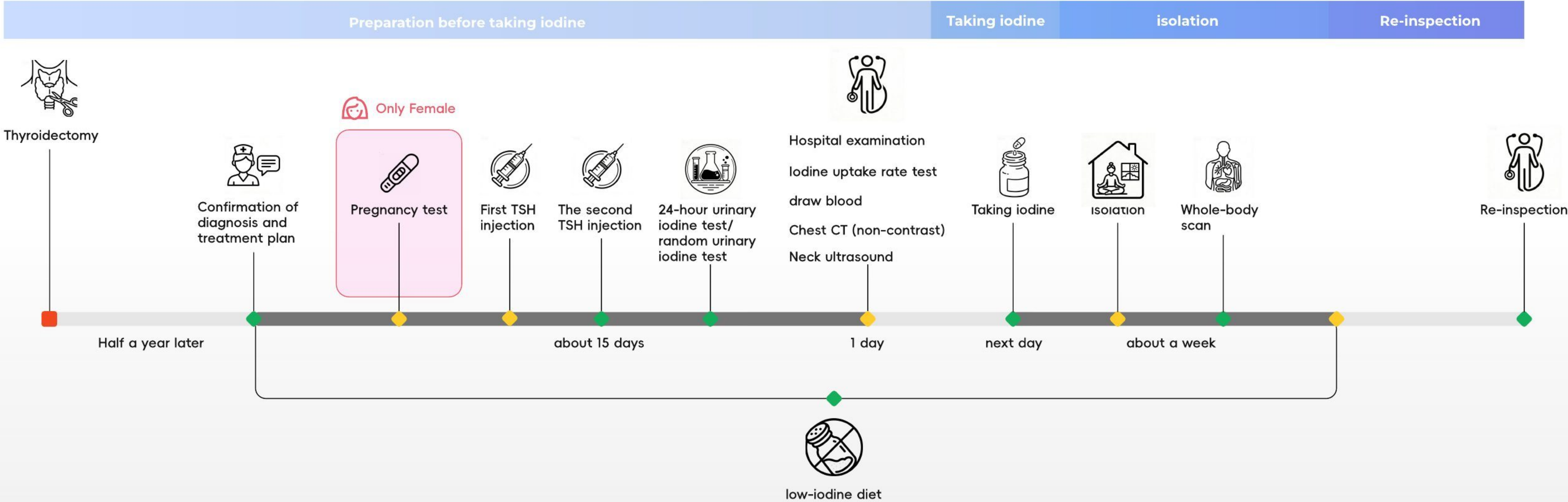
131 Treatment Processes



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Differences in I-131 Treatment Processes

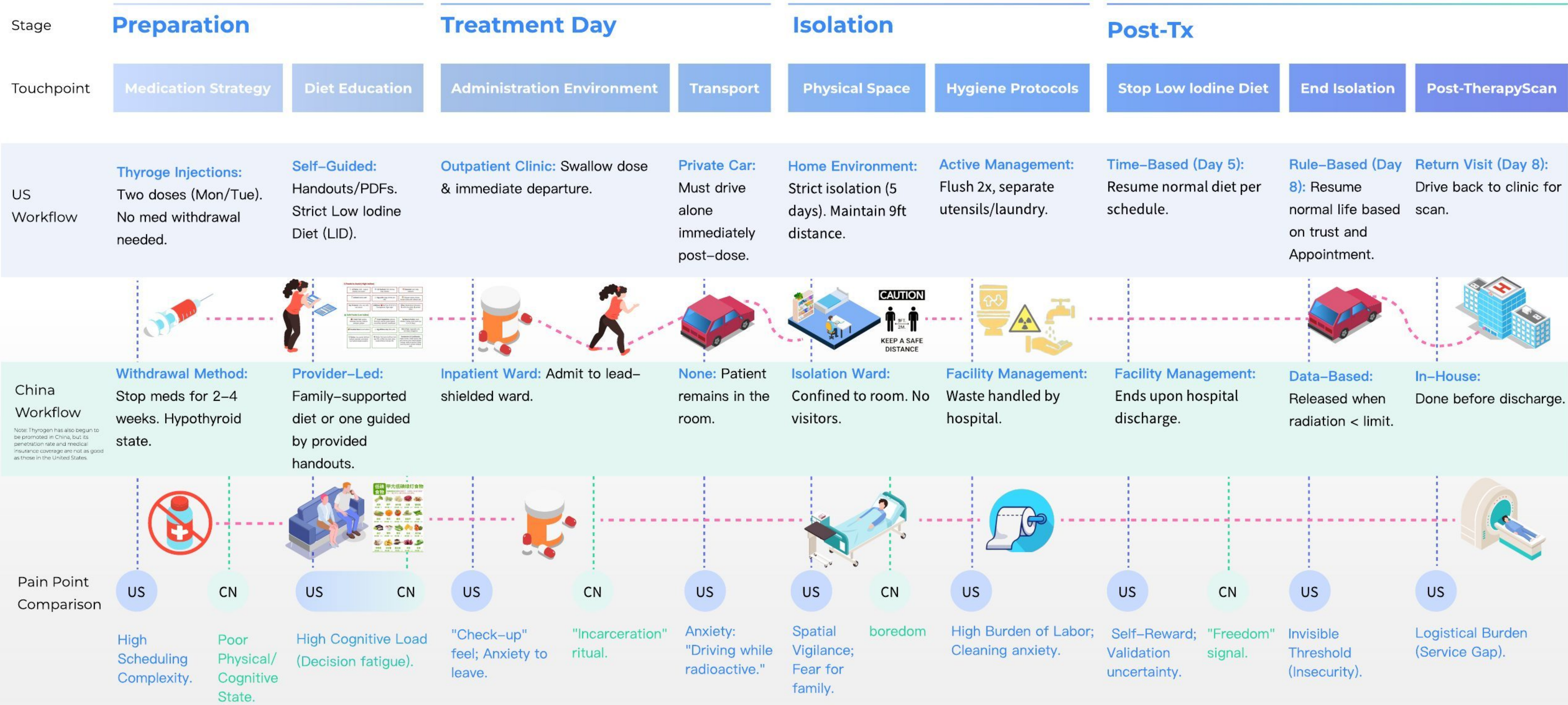
Between China and the United States



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ExploreSelf:

Fostering User-driven Exploration and Reflection on Personal Challenges with Adaptive Guidance by Large Language Models

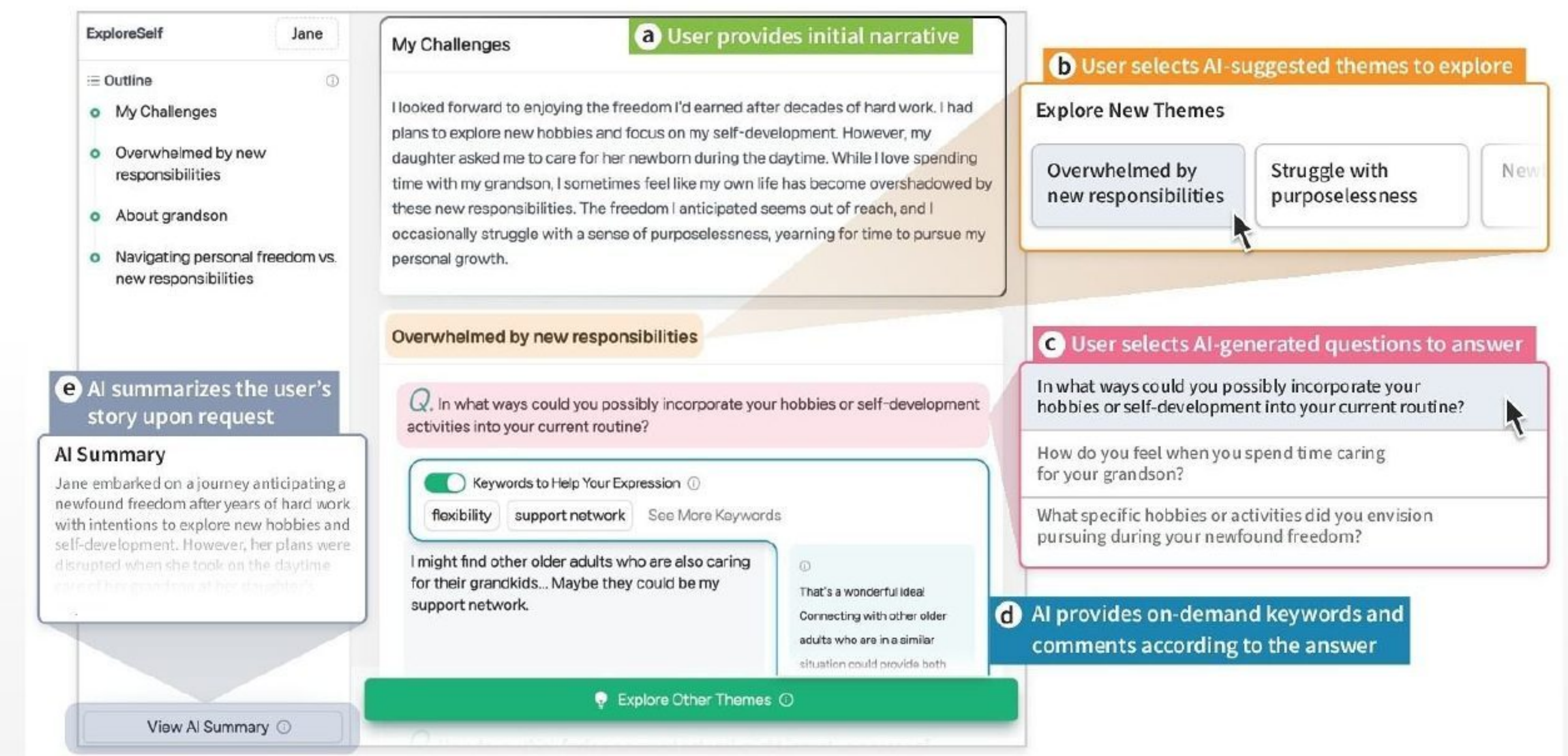
Research Insights

Expressing personal challenges such as stress and negative emotions through words **has been proven to improve physical and mental health.**

Existing reflection tools fail to balance the need for **‘providing sufficient guidance to users’** and **‘allowing users to control the reflection process’**, resulting in insufficient depth of reflection and low participation rates.

Core takeaway

The **Pathways Subscale** indicator measures the individual's ability to generate and maintain strategies for achieving goals
Provide key words and **immediate comments** to reduce emotional burden.



Exploring the Impact of

Avatar Representations in AI Chatbot Tutors on Learning Experiences

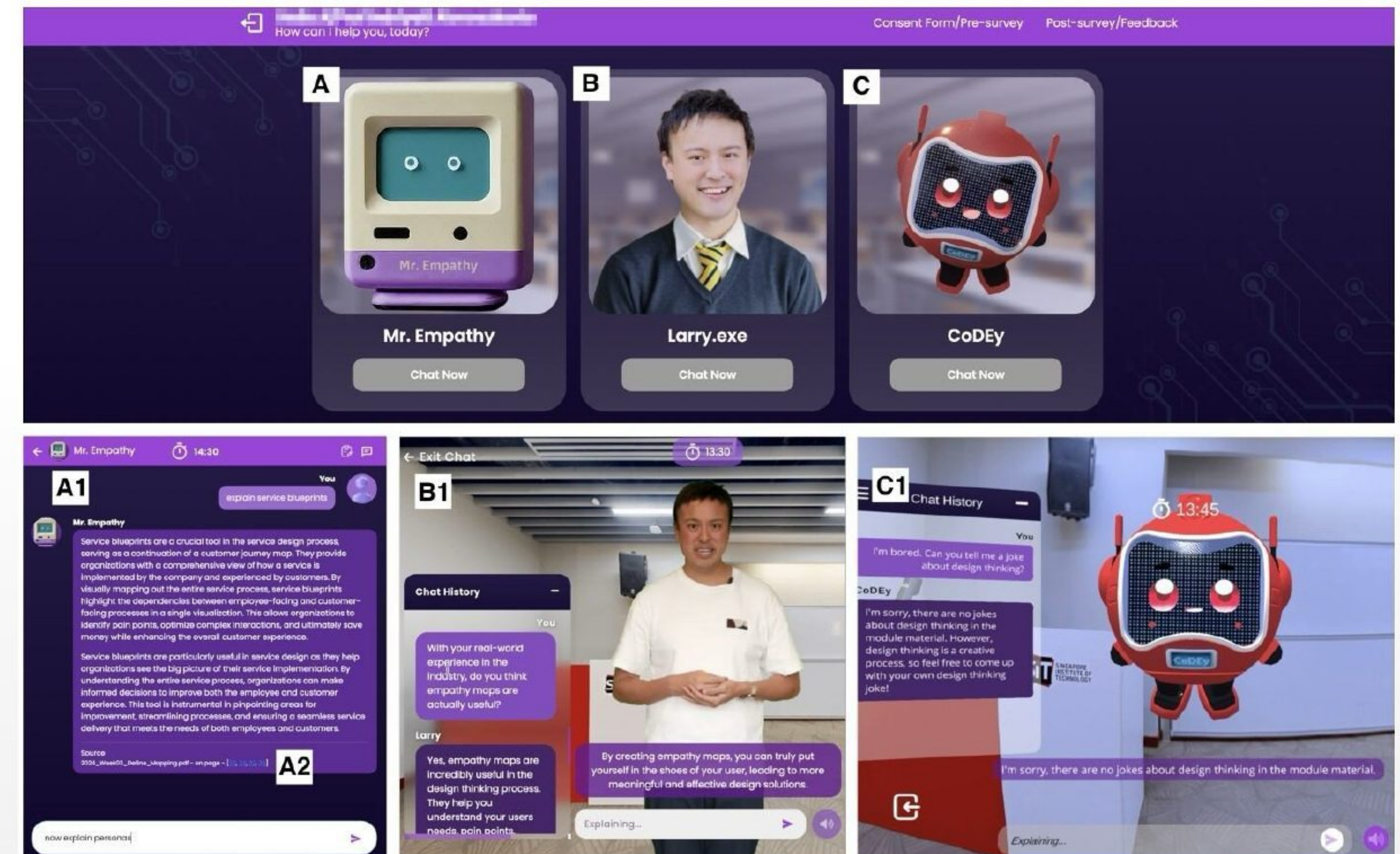
Research Insights

This paper explores the impact of different AI chatbot avatar representations on students' learning experiences through a **mixed-methods within-subjects study**.

Their findings reveal that preferences vary according to factors such as learning habits and learning activities. These insights suggest that the **thoughtful integration of diverse avatar representations in AI chatbot systems** for different learners and settings can greatly **enhance learning experiences**.

Core takeaway

Task-oriented learners prefer **text-based interfaces**, while **engagement-oriented** learners prefer **avatar-based interfaces**.
Provide key words and **immediate comments** to reduce emotional burden.



Towards Hormone Health:

An Autoethnography of Long-Term Holistic Tracking to Manage PCOS

Research Insights

Current tracking tools fail PCOS patients through false ovulation predictions and rigid designs. Effective management faces **medical** (interpretation gaps), **socio-cultural** (diet nuances), **temporal** (routine disruptions), **technical** (fragmented data), and **spatial** challenges.

Future tools must prioritize metabolic health over fertility, enable **granular hormone/diet tracking**, and **leverage smart home environments** for behavioral support.

Core takeaway

Designing for PCOS requires shifting from fertility-focused apps to **inclusive, holistic systems** that **integrate metabolic health**, **flexible multimodal inputs**, and **ambient smart home** cues to support long-term care.

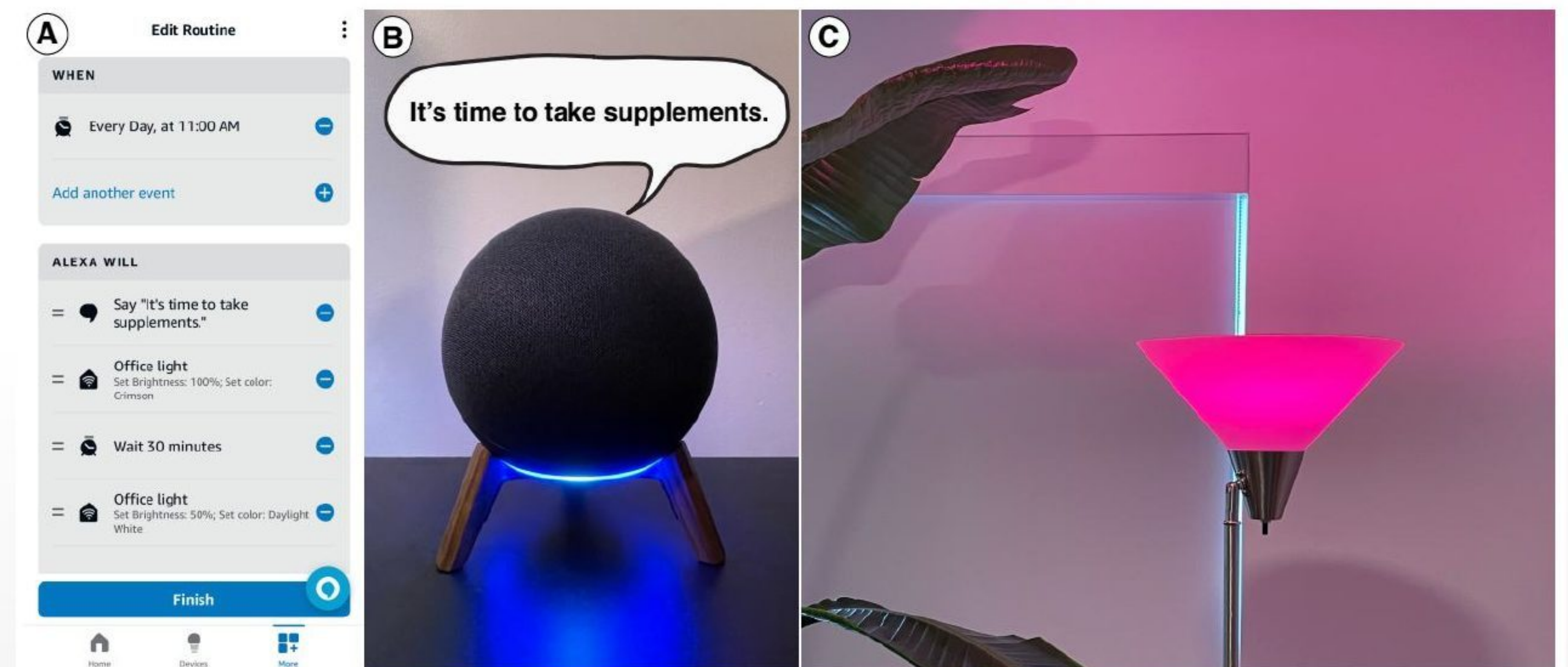


Figure 6: Smart home IoT Image ① displays a screenshot of the Amazon Alexa app, which the first author used to set up a daily routine for supplement intake reminders. Every morning at 11 a.m., the Amazon Echo Dot ② reminds her to take the supplement, and the floor light in her office changes to crimson (deep red color) for 30 minutes ③.

Related work

Exploring the Design Space of

Real-time LLM Knowledge Support Systems: A Case Study of Jargon Explanations

Research Insights

Expressing personal challenges such as stress and negative emotions through words **has been proven to improve physical and mental health.**

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Chemo

Definition:

A treatment method that uses powerful drugs to kill rapidly growing cancer cells.

Metaphor:

Chemo is like medicine soldiers fight cancer cells in a battle

List:

- **Related Concepts:** Cancer treatment, chemotherapy drugs, side effects
- **Uses:** Treating various types of cancer
- **Benefits:** Can shrink tumors, prevent spread
- **Limitations:** Side effects like nausea, hair loss

Image:



StopGap An LLM-powered Real-time Assistant for Knowledge Support

Video Pane

Choose File Talk_video.mp4 Play

Real-time Knowledge Support Pane

Flashcards

Induced Pluripotency

Pluripotent

Pluripotent is a term describing a cell capable of developing into any cell of the body (except for the placenta).

Induced Pluripotent Stem Cells or iPS cells are cells taken from a patient that are reprogrammed so that they can undergo differentiation into any type of cell in the body

Support formats vary across: text definitions, metaphors, images, lists

Foody Talk:

Exploring Opportunities for Conversational Food Journaling

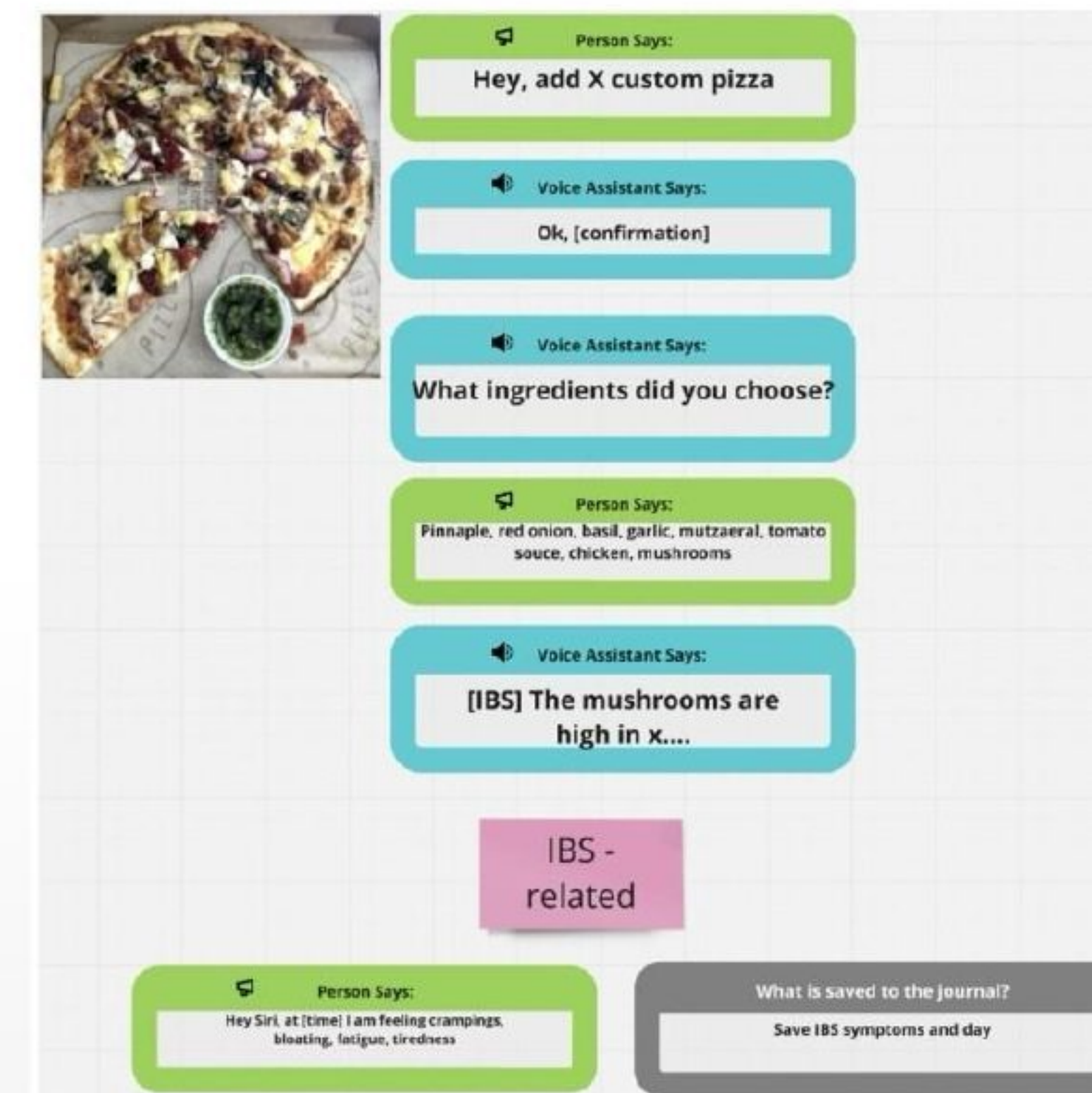
Research Insights

Expressing personal challenges such as stress and negative emotions through words **has been proven to improve physical and mental health.**

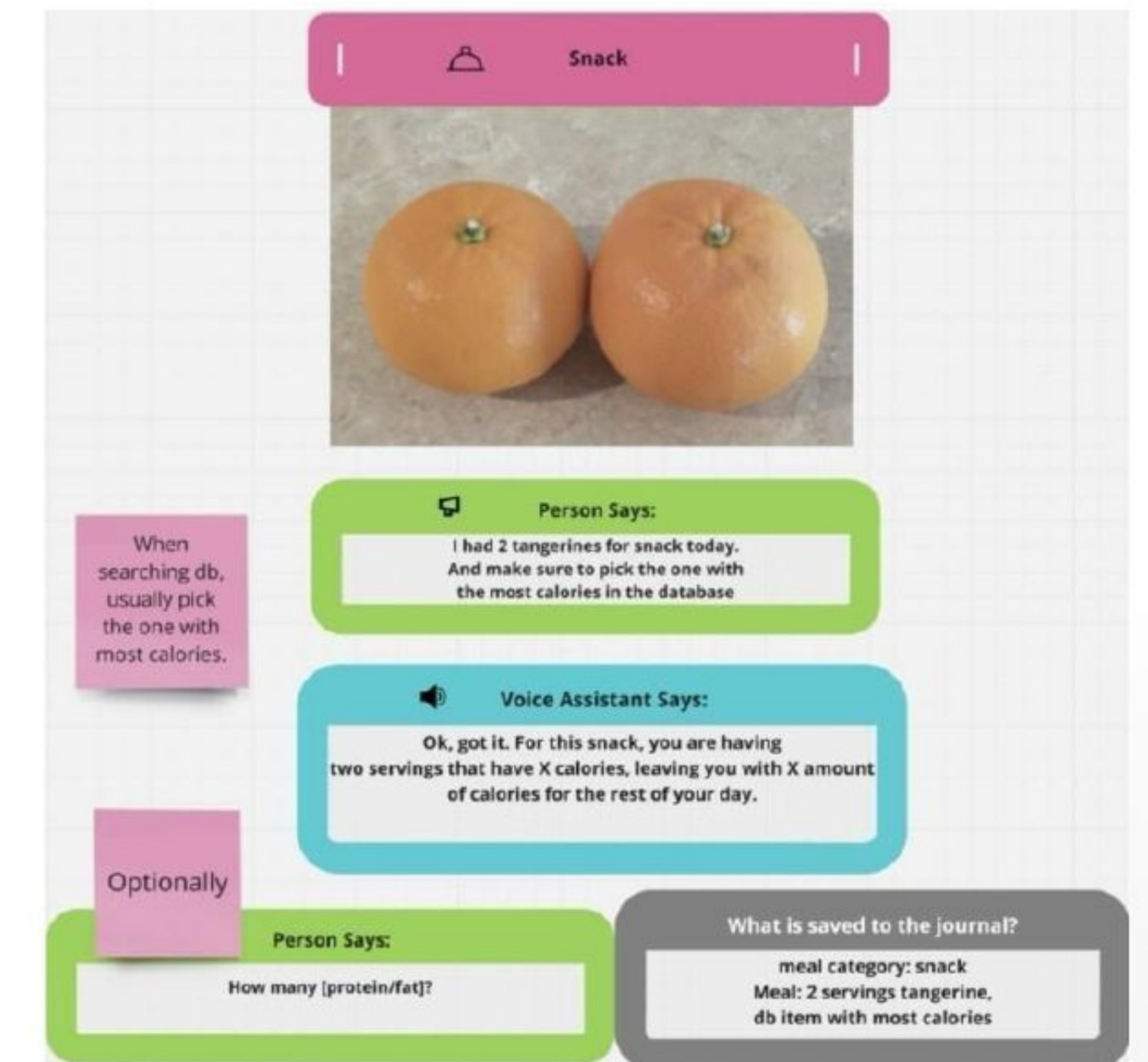
Existing reflection tools fail to balance the need for ‘**providing sufficient guidance to users**’ and ‘**allowing users to control the reflection process**’, resulting in insufficient depth of reflection and low participation rates.

Core takeaway

Provide key words and **immediate comments** to reduce emotional burden.



(a) P04 envisioned CUIs providing IBS-related food ingredient information and recording discomfort levels, supporting her condition management goal.



(b) P15 envisioned delegating nutritional database searches to CUIs while specifying a rule to select the one with highest calorie count, illustrating a desire for automation guided by preferences.

"I use video calling in all areas of my life":

Understanding the Video Calling Experiences of Chronically Ill People

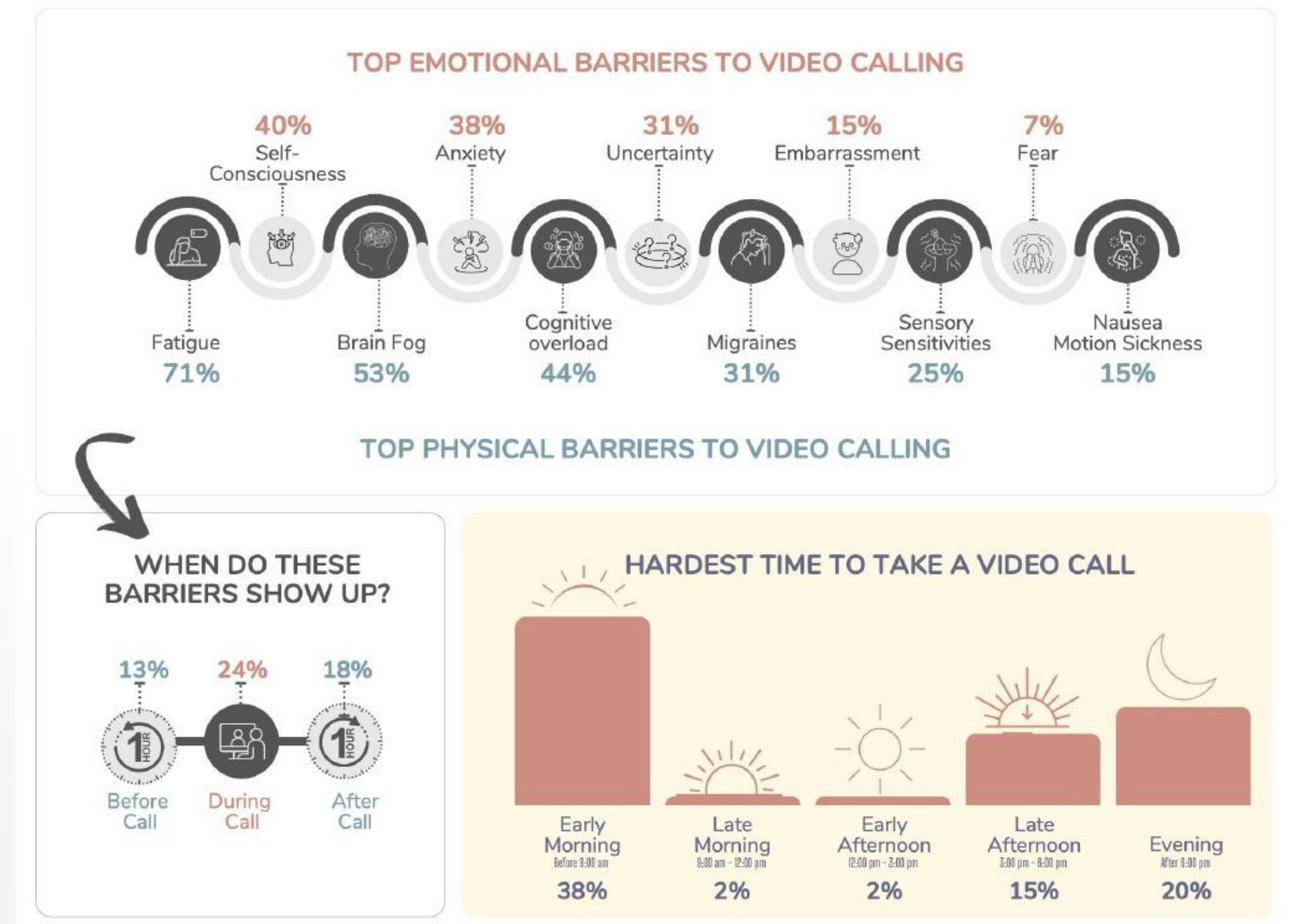
Research Insights

The Hidden Struggle Chronically ill users practice "**white knuckling**," expending energy to **mask symptoms** for productivity. They utilize an **off-camera ecosystem**, hiding medications and coping tools in **blind spots** to manage fluctuating conditions unseen by colleagues.

Vital but Rigid Video calling is a **vital lifeline** for safety and employment, not just a tool. However, current **static designs** fail to accommodate **unpredictable symptom fluctuations**, such as sudden **brain fog** or migraines, creating significant **cognitive barriers**.

Core takeaway

Leverage "**Aligned AI**" agents acting as **digital twins** to manage symptom fluctuations, providing **adaptive interfaces** and **non-verbal signals** to communicate hidden health states.



Related work



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Core takeaway

Understanding the Video Calling Experiences of Chronically Ill People



From "active input" to
**"Passive/natural
capture"**



Contextual and lightweight
**Just-in-Time
Support**

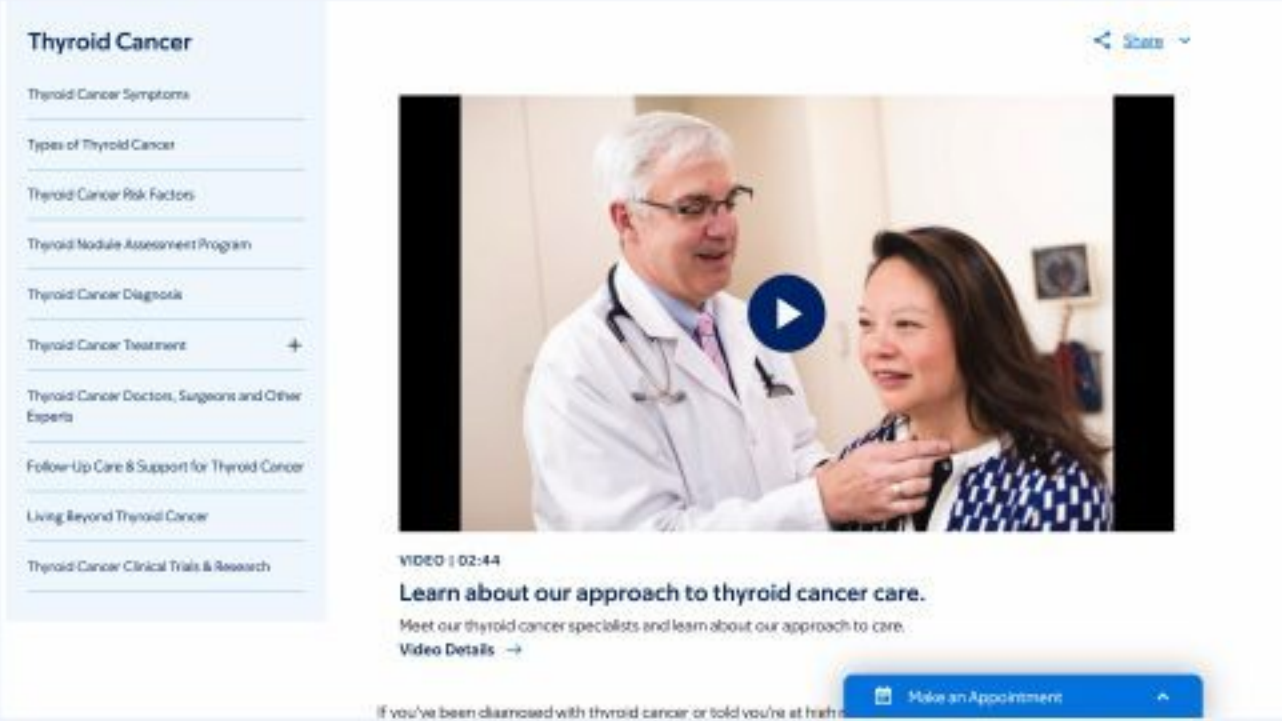
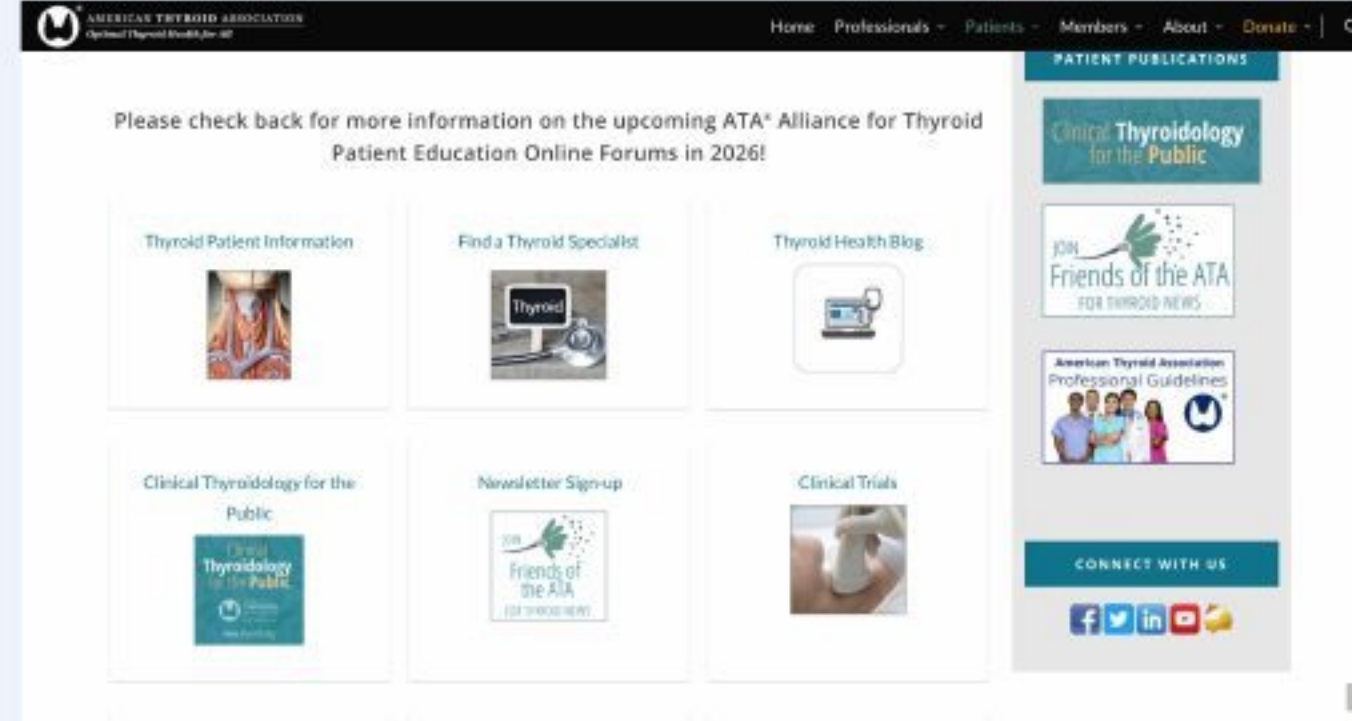
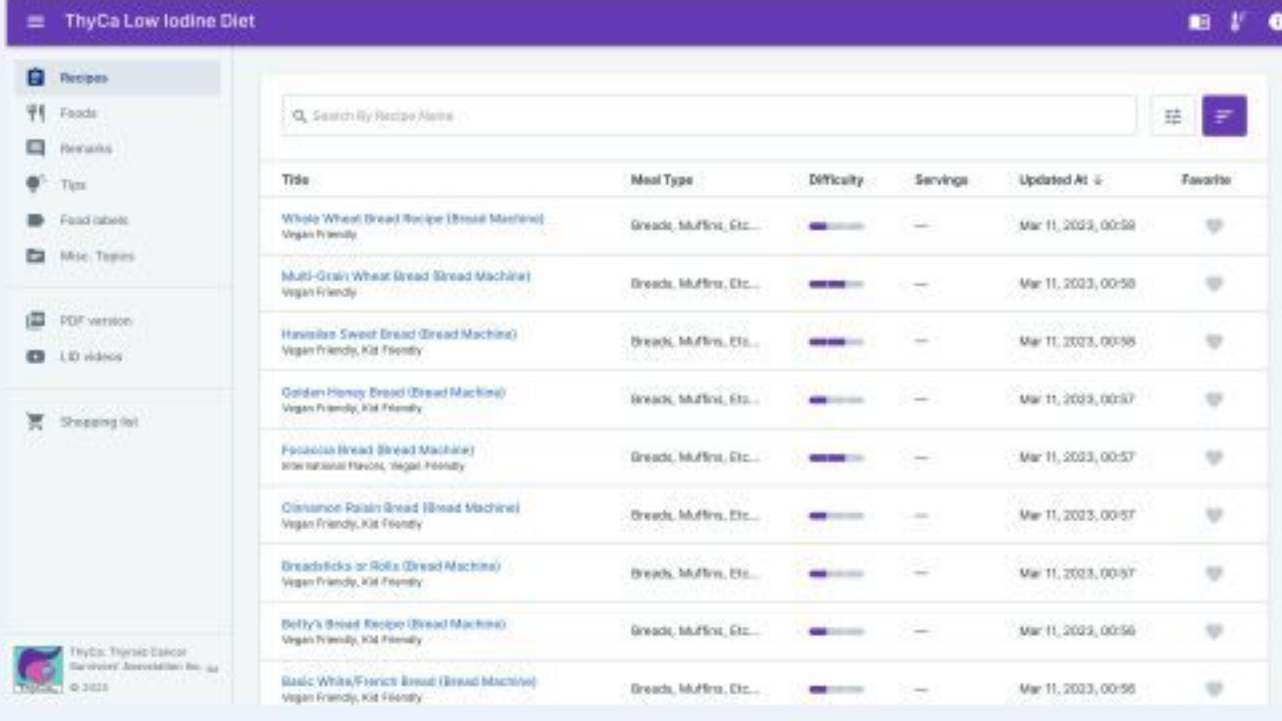


pure text form to
**Multimodal
expression**

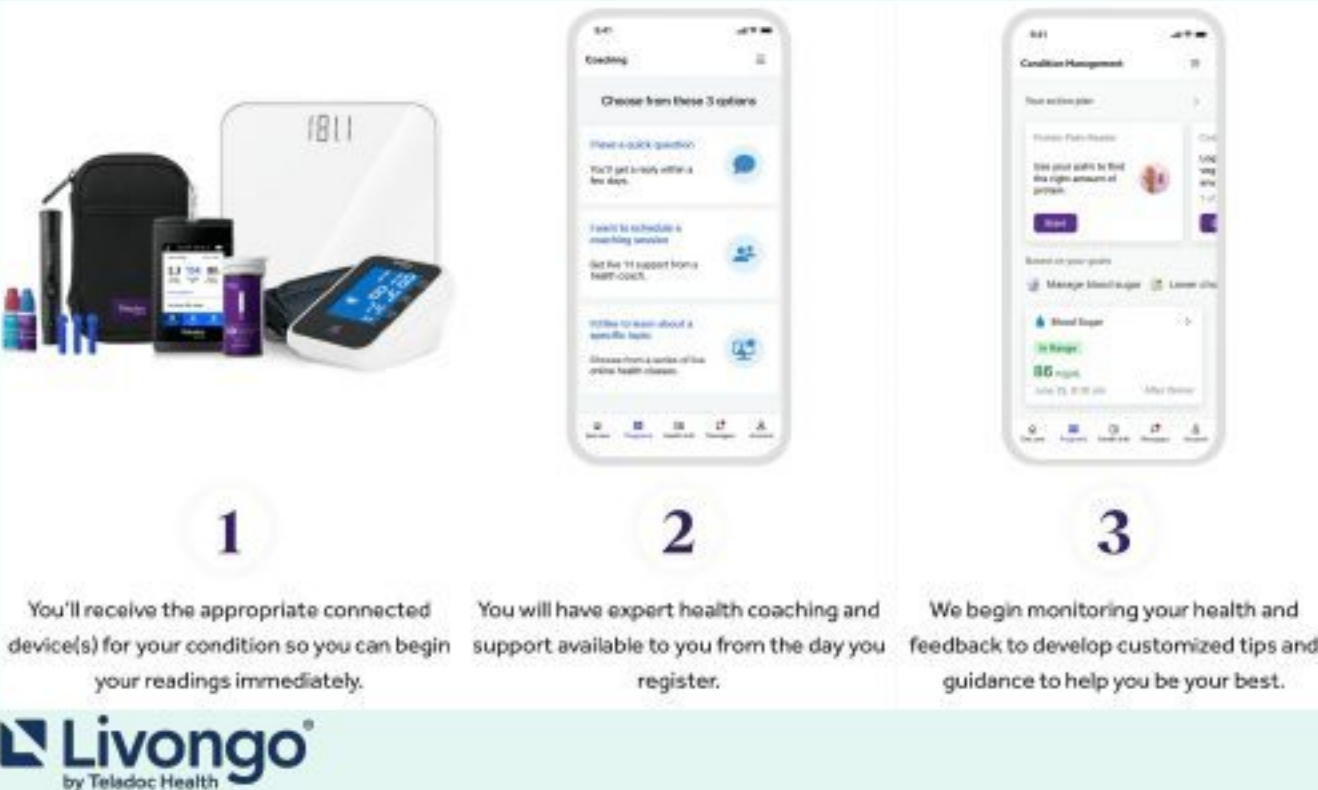
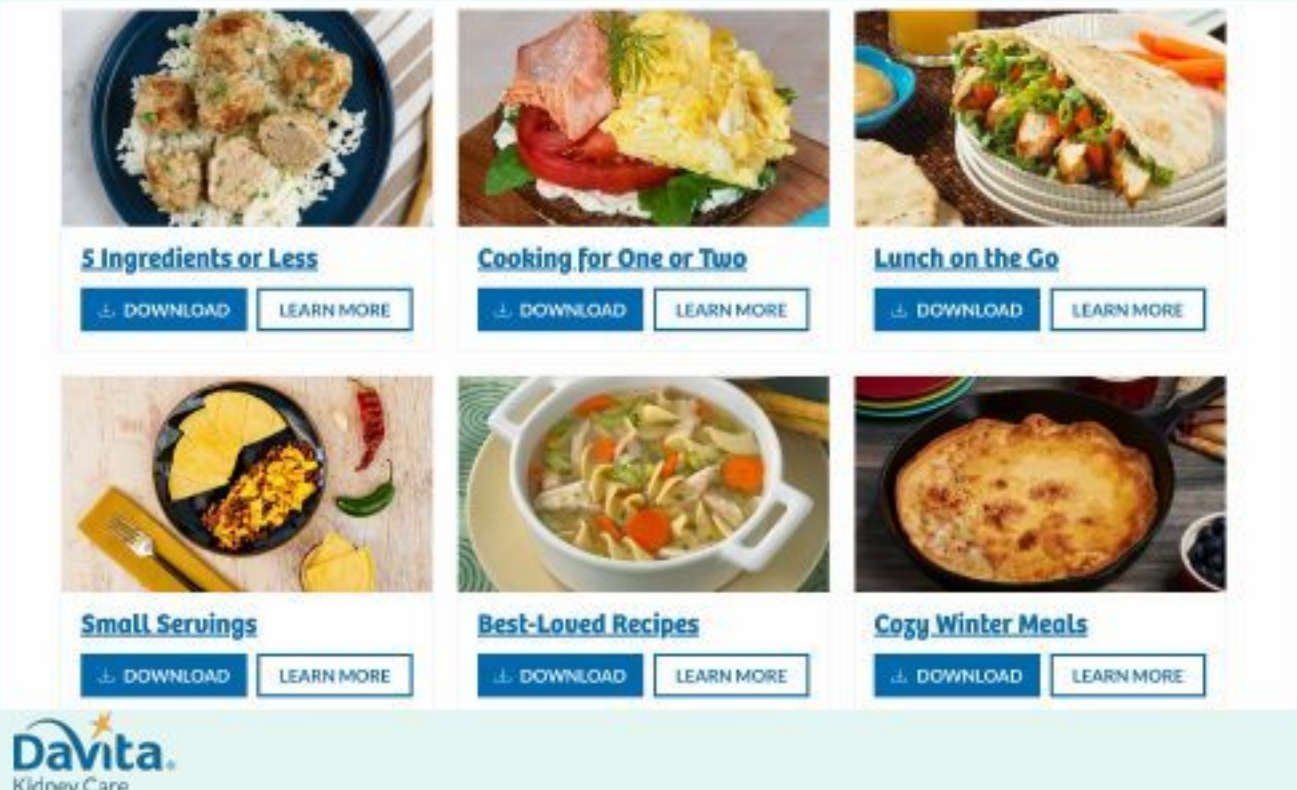


Give users sense of
**Control + Privacy
security**

Competitive Products

Product	Epic MyChart  Web/App	MSKCC RAI Patient Education  Web/Pdf	American Thyroid Association  Web/Pdf	ThyCa-Low Iodine Diet  Web/App
Type	Hospital patient portal	Center-based education	Authoritative medical website	Patient advocacy diet tool
Visualization	Basic lists & icons	Text-based	Minimal visuals	Checklist-based
Comprehensibility	Clinically worded	Clear, authoritative	Accurate but abstract	Patient-friendly
Timeline Clarity	Task-based, not holistic	Clear day-by-day	No timeline	Diet-only
Home Isolation Support	No spatial/family support	No home execution support	No practical isolation guidance	No isolation support
Personalization	Limited to orders	None	None	Recipe-level
Prons	● Compliant, scalable ● integrated	● Well-structured ● trusted	● Highly trusted	● Strong focus ● trusted
Cons	● System-centric ● low experiential support	● Non-interactive	● Informative, not actionable	● Fragmented from care journey

Functional Alternatives Analysis for Other Diseases

Product	Livongo 	mySugr 	Ovia 	DaVita Care 
Product Positioning	Long-term self-management	Everyday self-tracking	Life-stage & reproductive health	Patient advocacy diet tool
Core Strength	Adherence-focused coaching	Low-friction logging	Stage-based journey design	Checklist-based
Timeline / Care Path	Implicit longitudinal rhythm	No care pathway	Week-by-week timeline	Patient-friendly
Diet-related Features	High-level guidance	Food logging as core	Supportive guidance	Diet-only
Visualization & Comprehensibility	Clear trend-based feedback	Friendly, playful UI	Strong temporal visualization	No isolation support
Personalization	Data-driven personalization	Input-based	State-based personalization	Recipe-level
Prons	- Proven long-term adherence - Scalable behavior change	- Very accessible UX - High engagement	- Excellent expectation setting - Strong personalization	- Robust compliance support - Clear behavioral rules
Cons	- No constrained-living support - No explicit care timeline	- Log-centric - No journey orchestration	- Not a clinical workflow - Limited constraint handling	- Limited emotional design - Tool-oriented UX

Research Objectives

Identify core pain points for patients across all phases of I-131 therapy

With particular focus on low-iodine diet compliance, home isolation practices, scheduling, and emotional management.

Evaluate the effectiveness of existing educational materials

Assess whether current channels (e.g., printed guides, phone support, patient portals) meet real-world needs and identify gaps or contradictions in information.

Understand care team workload and recurring issues

Determine which questions patients ask most frequently and which instructions are commonly misunderstood, to pinpoint opportunities for automation or simplification.

Explore opportunities for digital interventions

Identify which aspects of the process would benefit most from visual, interactive, or intelligent tools (e.g., mobile apps, checklists, videos, AI-powered Q&A) to enhance patient autonomy and experience.

Identify characteristics of high-risk or high-difficulty patient groups

Such as single parents, individuals living in small spaces, older adults, or those from cultural backgrounds that emphasize family caregiving—to inform tailored support strategies.

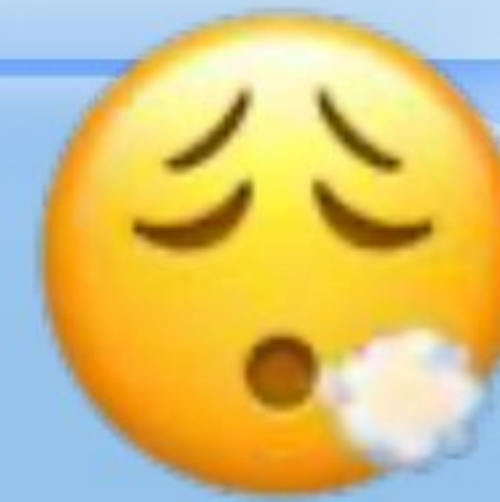
Patient Perspective

I'd get confused and then I'd panic.



I tried looking things up on Google a few times but that **made it WORSE** - I'd read something on some forum where someone said they did this or that, and then I'd wonder **if my doctor wanted me to do it that way too.**

Honestly I don't really know how to use all that stuff on the computer.



My daughter helped me look at MyChart but honestly I don't really know how to use. The nurse was really my lifeline. **I probably called them 10 times that week.**

EVERYTHING about that diet confused me



Was I supposed to buy those? Could I use any salt at all? **I was too scared to eat ANYWHERE.** I was terrified of messing it up.

Care Team Perspective

Diet - patients call multiple times asking about foods



They need a lot of reassurance that their experience is typical. Many patients call multiple times to **verify the timeline** because they're anxious about getting it wrong.

Single parents or primary caregivers have the most difficulty



especially with young children who don't understand why they can't be close to mom or dad. These patients often need **the most coaching** and sometimes compromise safety out of practical necessity.

WHAT...IF...



Our handouts are generic rules without **practical problem-solving**. And the emotional and psychological aspects aren't addressed.

Key Insight

- Patients struggle with ambiguous ‘low-iodine diet’ guidelines, especially regarding **food label ingredients , salt types, and dining out.**
- Current materials lack **positive guidance.**
- The ‘9-feet distance’ rule is nearly impossible to maintain—leading to **emotional distress and guilt.**
- There’ s a lack of **contextual visual aids**, forcing patients to guess.
- Written materials are overly technical and static; patients **struggle to absorb them while anxious.**
- Conflicting advice **across sources** increases confusion.
- Diet and isolation questions account for **over 40% of phone calls.**
- Nurses express a strong desire to offload “basic FAQs” so they can focus on true medical issues.
- Patients commonly experience loneliness, fear, yet **no emotional support** in the care pathway.
- Care teams empathize but lack time to address non-medical concerns.

01

Diet management is the biggest barrier—and heavily reliant on external validation

02

Home isolation rules are disconnected from real-life contexts and hard to follow

03

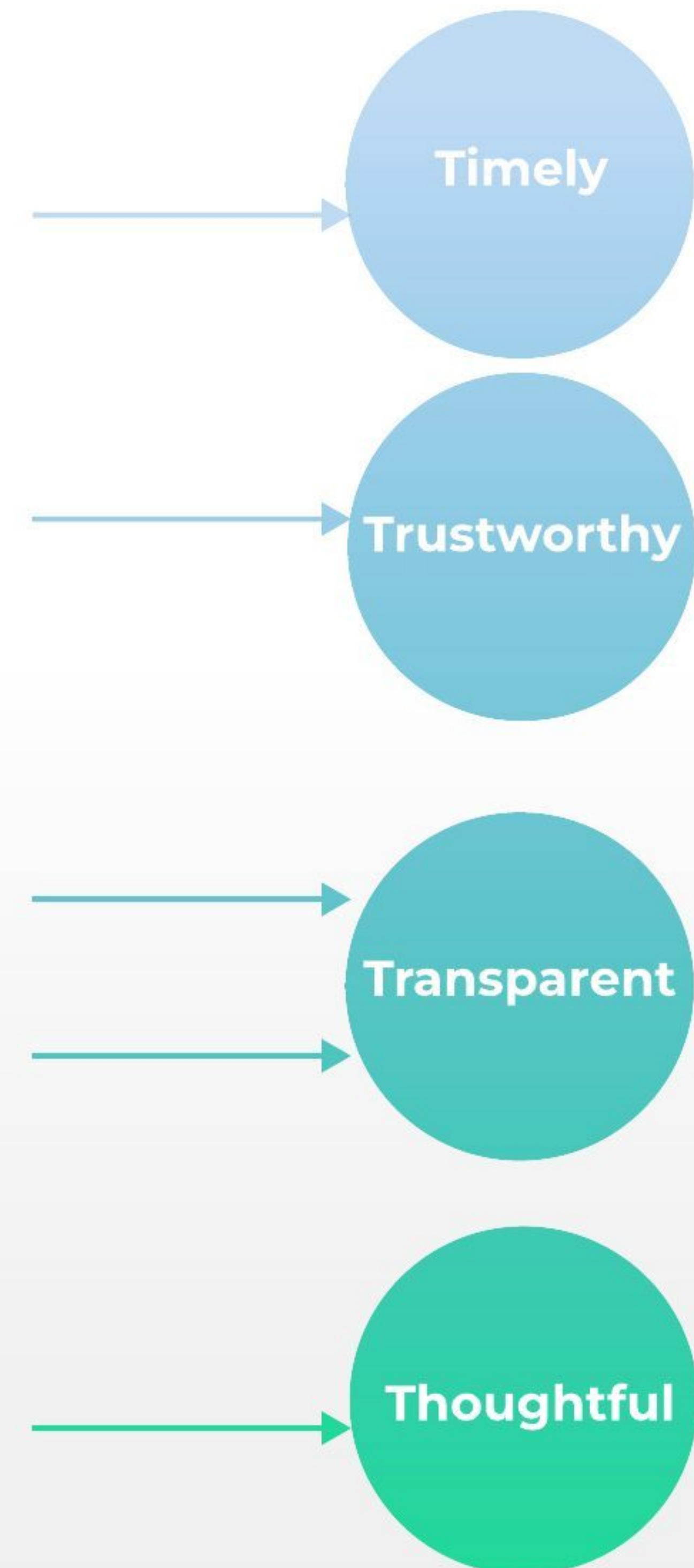
Information delivery doesn’ t match patient cognition under stress

04

Care teams are overwhelmed by repetitive, low-complexity inquiries

05

Emotional and psychological support is severely lacking



Design Vision

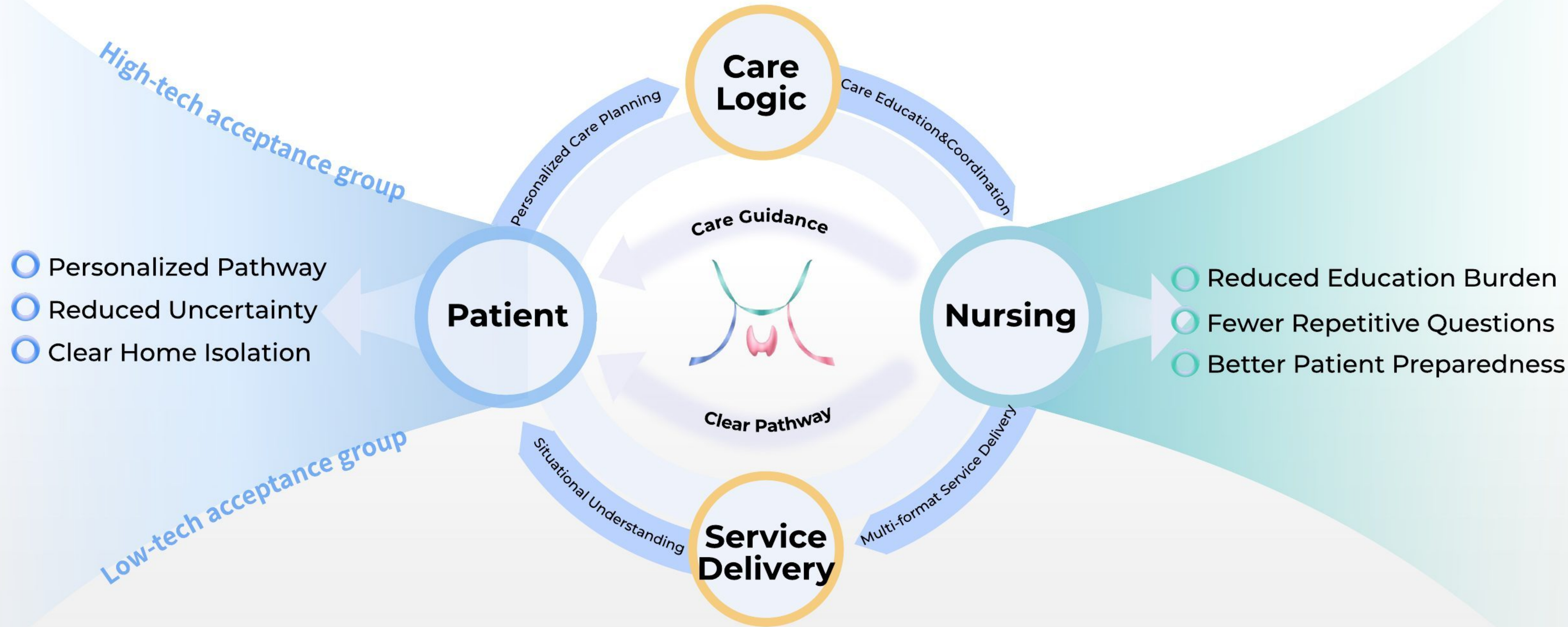


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Transform I-131 treatment from a complex, uncertain process into a clear, understandable, and actionable care experience.



From standardized instructions → personalized, situation-aware care pathways.

Offering Map



Core Offering

Level 1

Low-Iodine Diet Enablement

Nurse Workload Offloading

Home Isolation Support

Anxiety & Uncertainty Reduction

Personalized Care Path Generation

Patients

Clear and predictable treatment journey
Reduced anxiety and risk of mistakes
Aligned with real-life living conditions



Decision Support

Level 2

Low-Iodine Diet Voice Logging

Diet Double Confirmation

Context-aware Daily Decisions

Schedule Guardian

Nurses

Reduced repetitive education
More time for high-value care
Lower communication and coordination burden



Reassurance & Care

Level 3

Home Isolation Visualization

Radiation Decay Over Time

Emotional Normalization Prompts

Family-Facing Summary View

Healthcare System

Improved treatment adherence
Reduced procedural errors
Increased patient satisfaction



The chatbox serves as the patient's decision interface, while the agent system ensures context-aware, stage-specific guidance across the treatment timeline.

New Service Flow



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Service
Stages

Pre-treatment

Low-Iodine Diet

RAI Day

Home Isolation

Stop Diet

End Isolation

Post-tx scan

Recovery

Understand the overall plan
and what to prepare first

Choose treatment-safe foods
before eating

Same-day tests &
meds in order

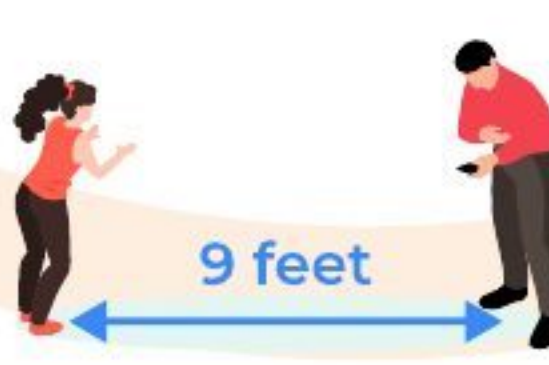
Follow radiation safety rules
to protect others

Confirm diet ends;
isolation continues

Confirm isolation end;
contact safe

Understand scan
purpose

Understand long-
term follow-up care



Patient
points

- At home;
- printed schedule
- MyChart

- Kitchen
- grocery store
- eating out

- Clinic
- Nuc Med

- bedroom
- bathroom

- home

- Home → daily activities

- Home → hospital

- Daily life

Patient
Action

- Reviews schedule
- confirms outpatient or inpatient

- Checks food
- avoids guesswork

- Follows prompts
- checks next

- Follows isolation guidance daily

- Confirms rule change

- Confirms before resuming activities

- Schedules post-therapy scan
- Returns for scan

- Attends follow-ups

Interface
Layer

- Timeline overview
- Chatbox 「What should I do now?」

- LID explainer video
- Chatbox 「food check」

- Mobile alerts
- day-of checklist
- chatbox 「next step」

- Isolation visualization
- daily check entry

- Status update
- diet-end confirmation

- Isolation-end confirmation
- status update

- Scan reminders
- explanation

- Follow-up reminders
- question entry

Agent
Logic

- Detects care path
- Surfaces personal timeline

- Confirms diet phase
- Adapts guidance to context

- Tracks day-of progress
- Unlocks next

- Adapts guidance
- Adjusts reminder intensity

- Confirms rule timing

- Confirms isolation end timing

- Explains scan transitions flow

- Personalized monitoring

Nursing
Feedback

Fewer basic
process questions

Diet questions
largely resolved

Fewer step-
confirmation
questions

Reduced off-hour
anxiety contact

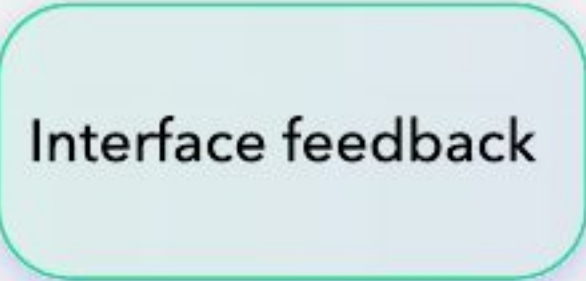
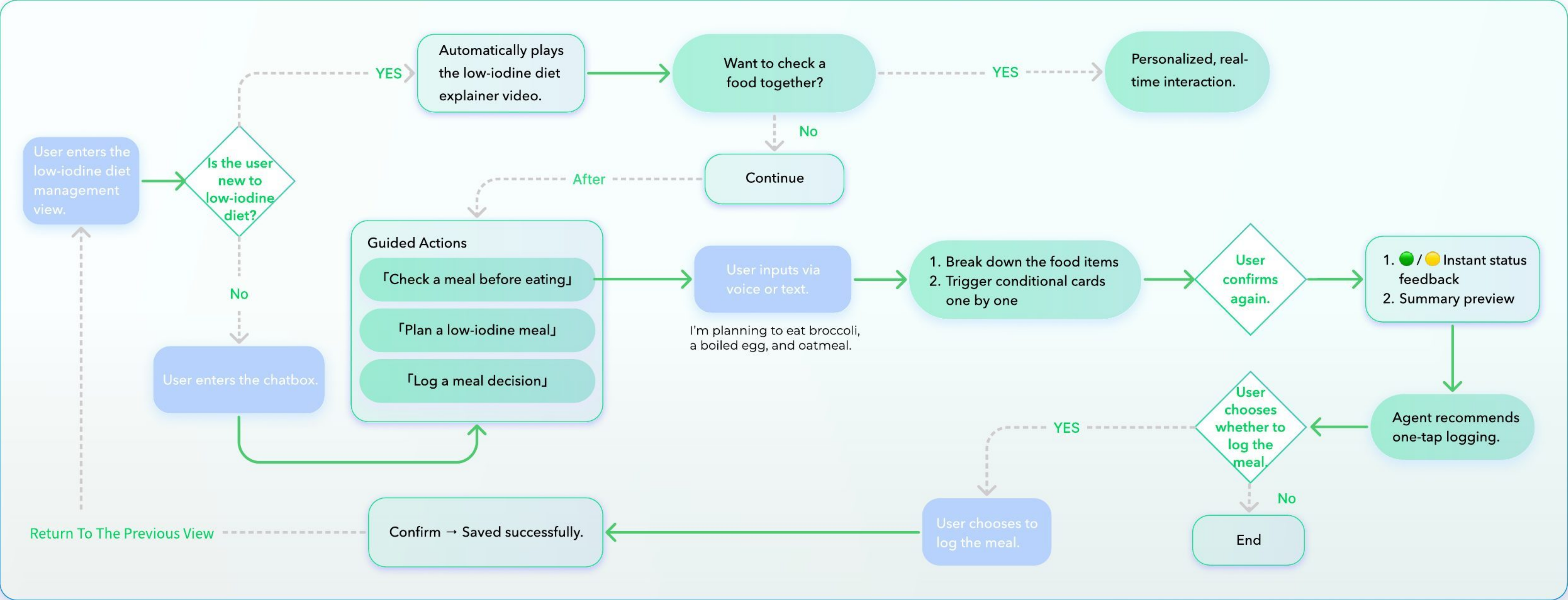
Fewer
clarification
questions

Reduced
premature
isolation ending

Better-
prepared
patients

Smoother
long-term
care

Interaction Flow—Low-iodine diet management



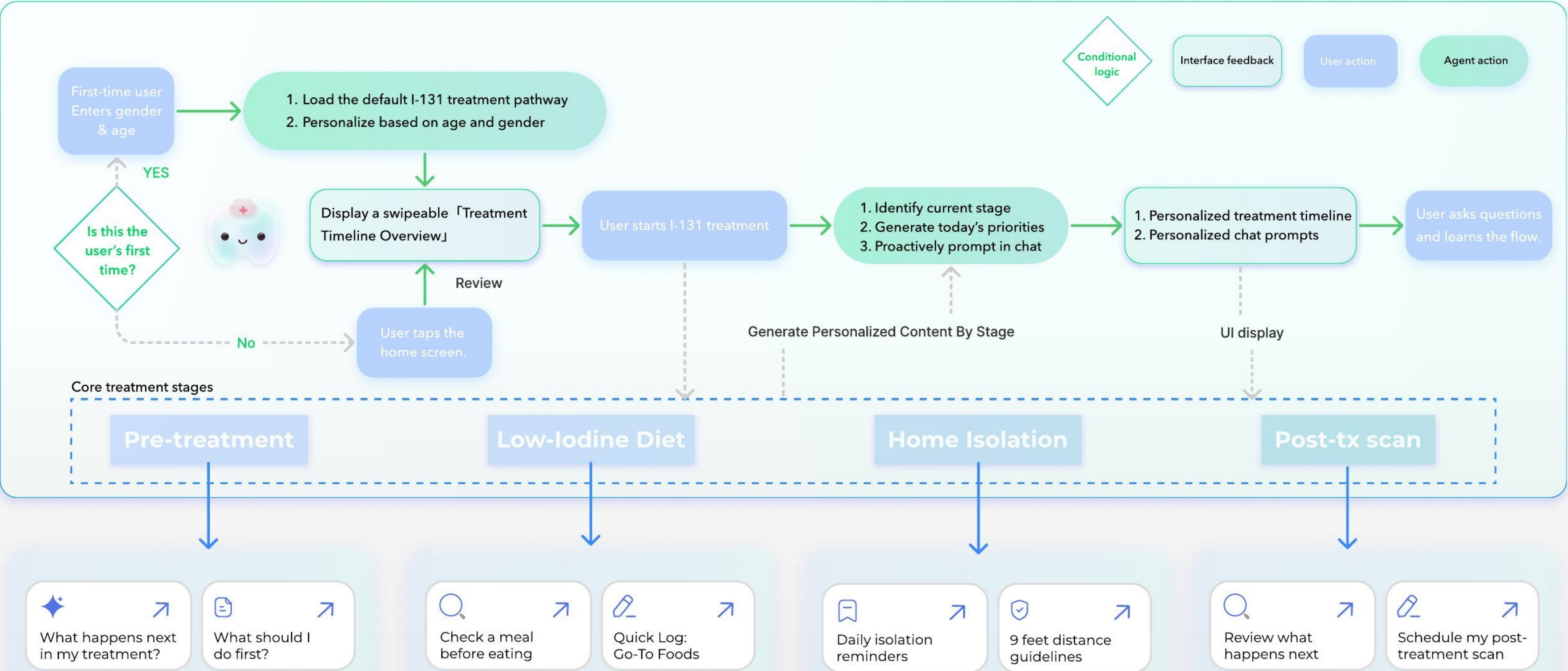
Interaction Flow—Personalized treatment timeline path



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Design output

Thyra



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Design Concept

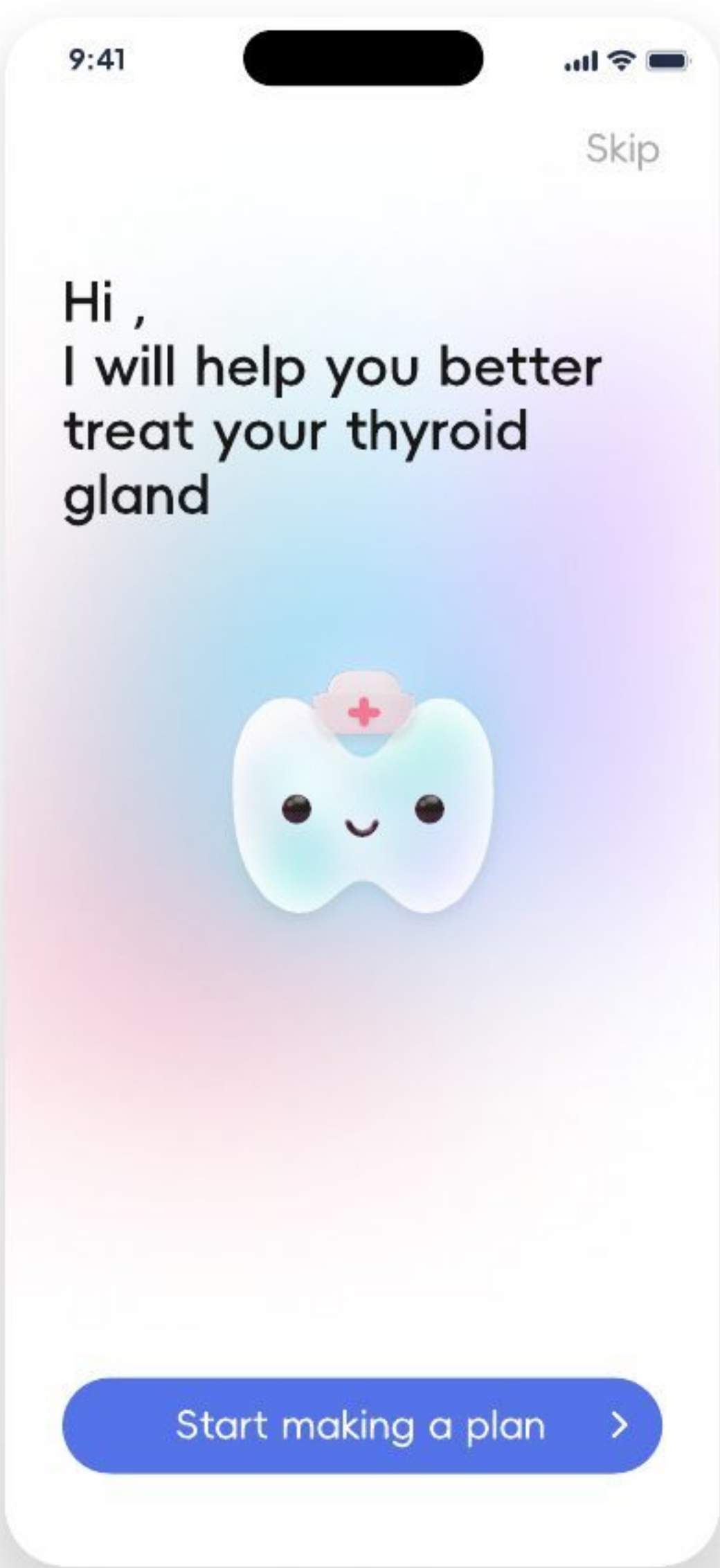
The thyroid, shaped like a butterfly, once guarded your body's rhythm.

When it's gone, Thyra 希芮 becomes your digital butterfly—gracefully guiding you through I-131 therapy with Trustworthy, Thoughtful, Timely service.

Thyra

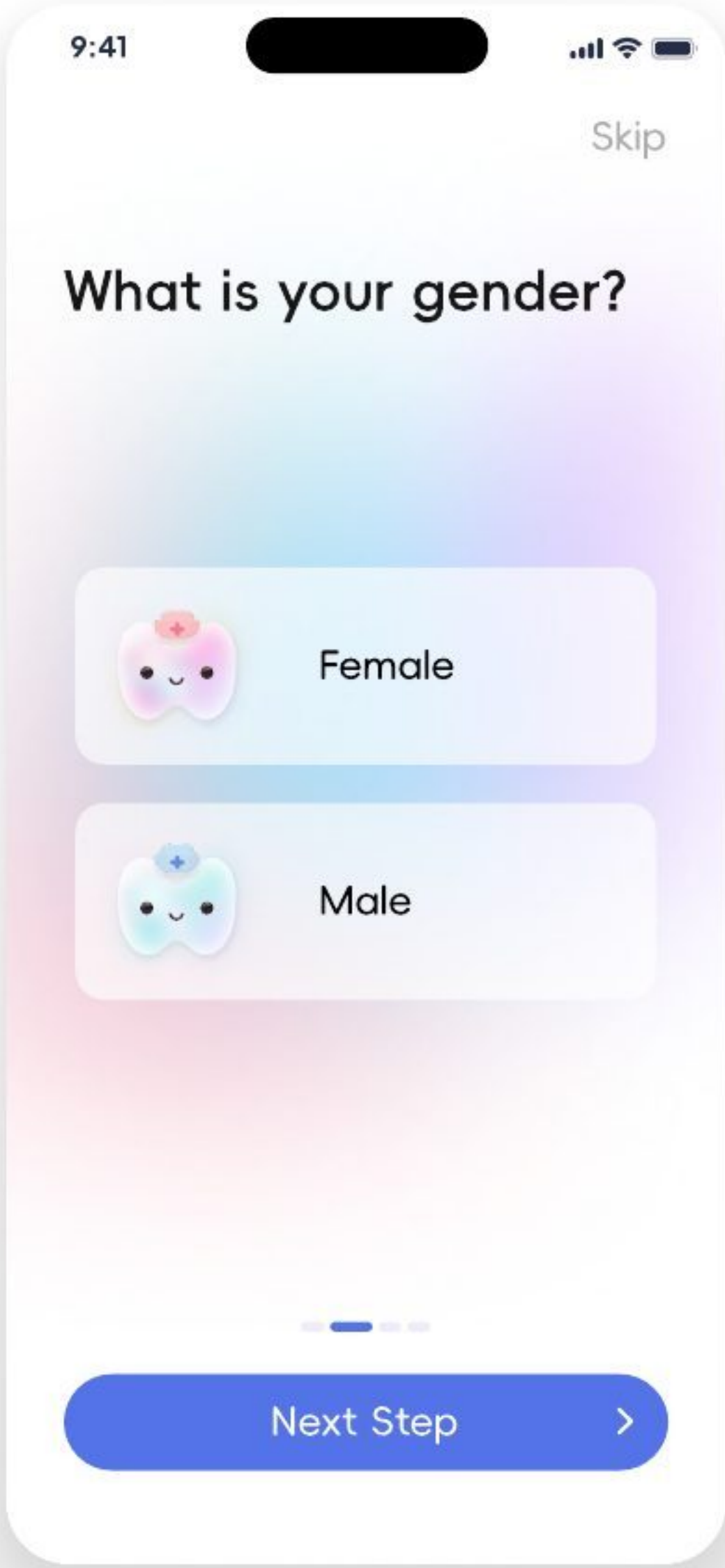
Trustworthy Thoughtful Timely

Personalized calendar planning



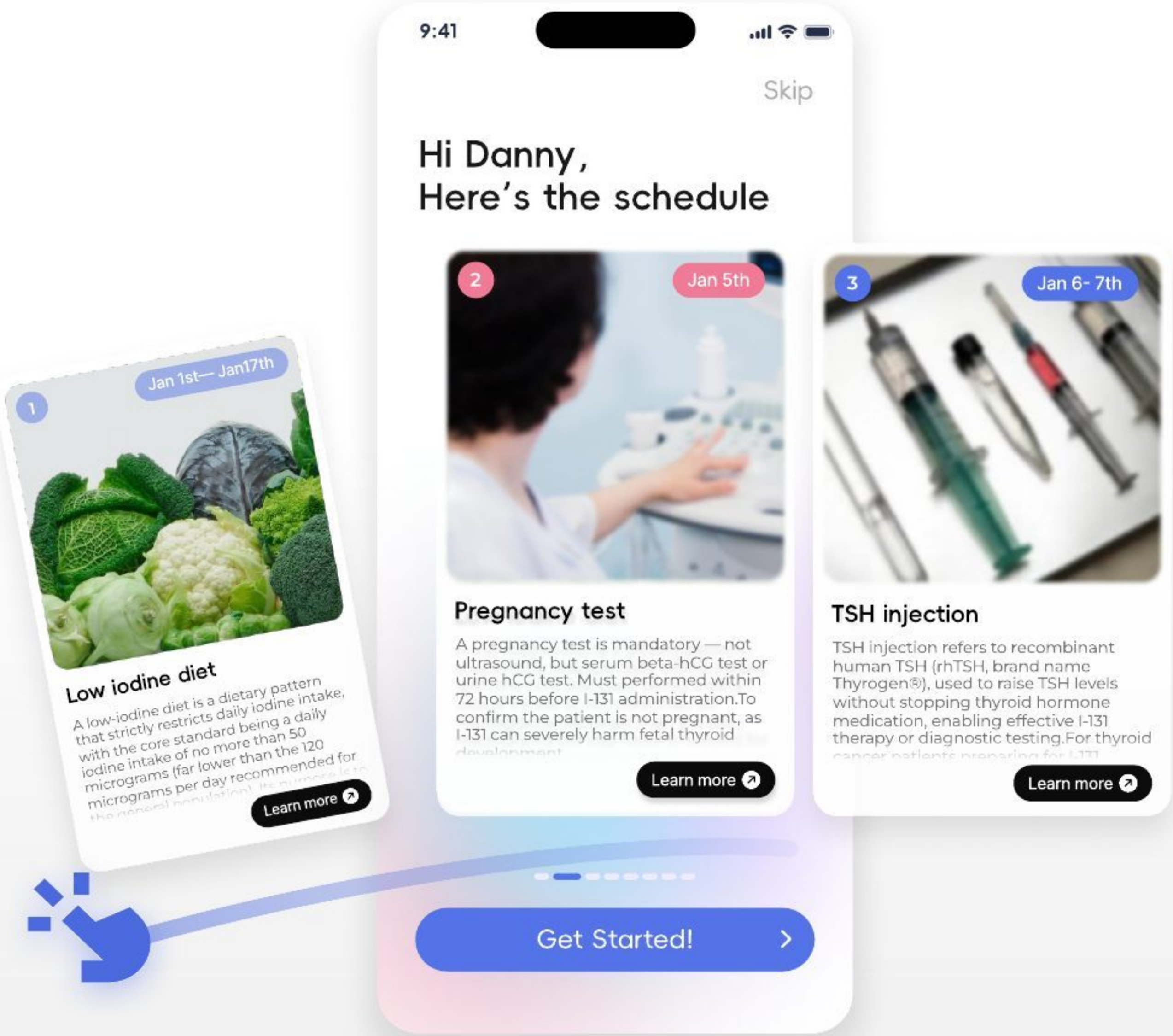
Welcome to Thyra 希芮

With thyra, easily start your I-131 treatment journey



Onboarding Q&A

A 5-step questionnaire accurately collects treatment information.



Pre-treatment health education card

Before generating a personalized calendar, knowledge cards will be used to introduce the entire process to the user, making it clearer.

Management of Diagnosis



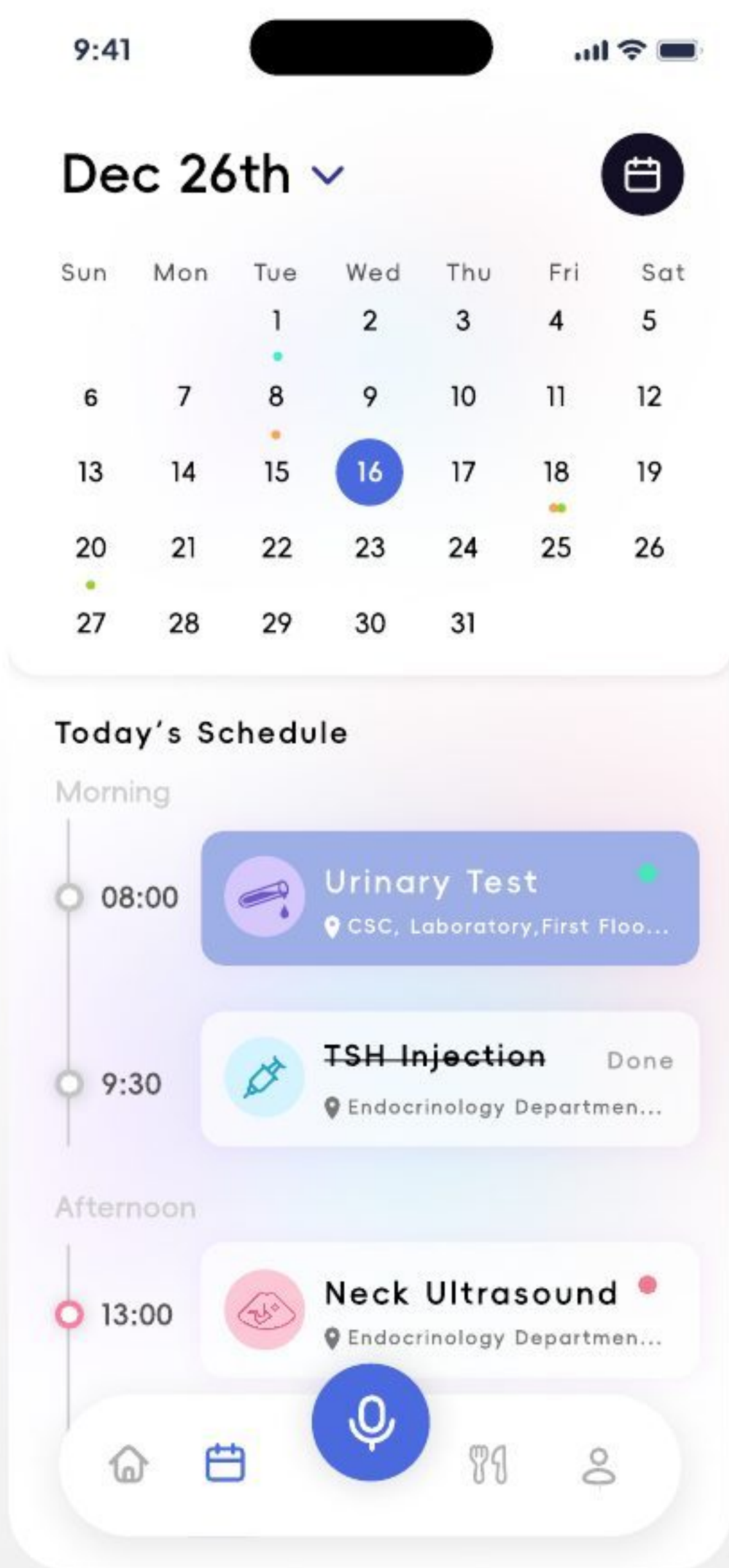
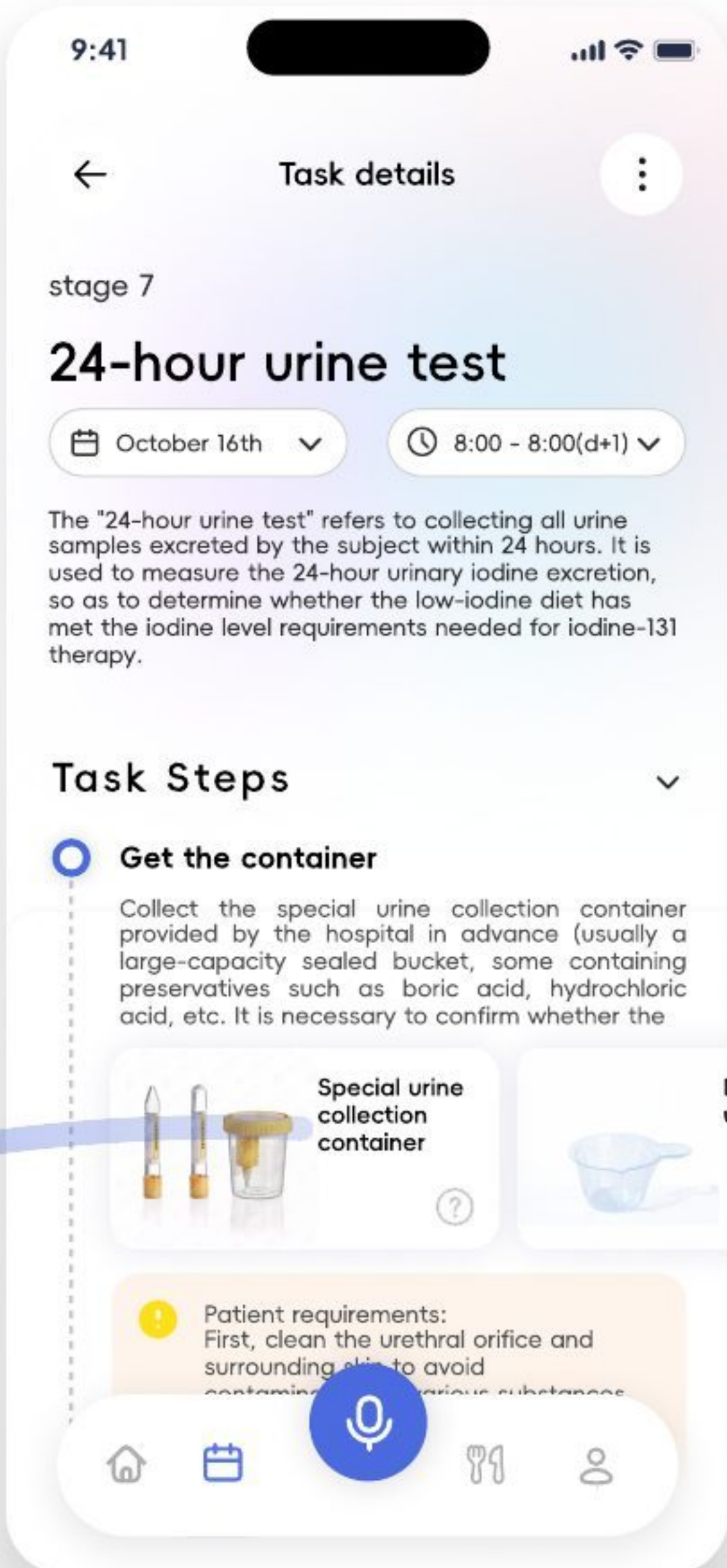
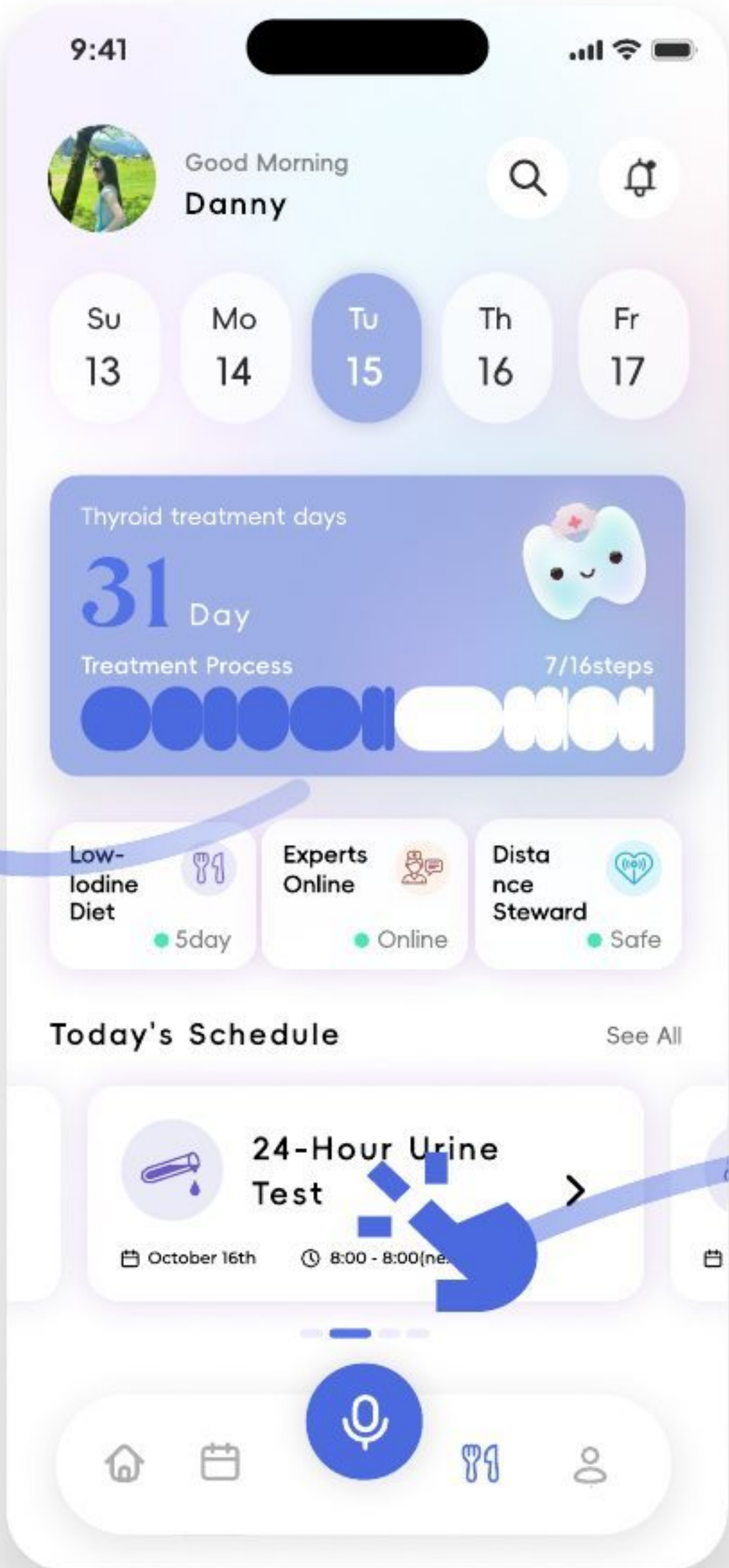
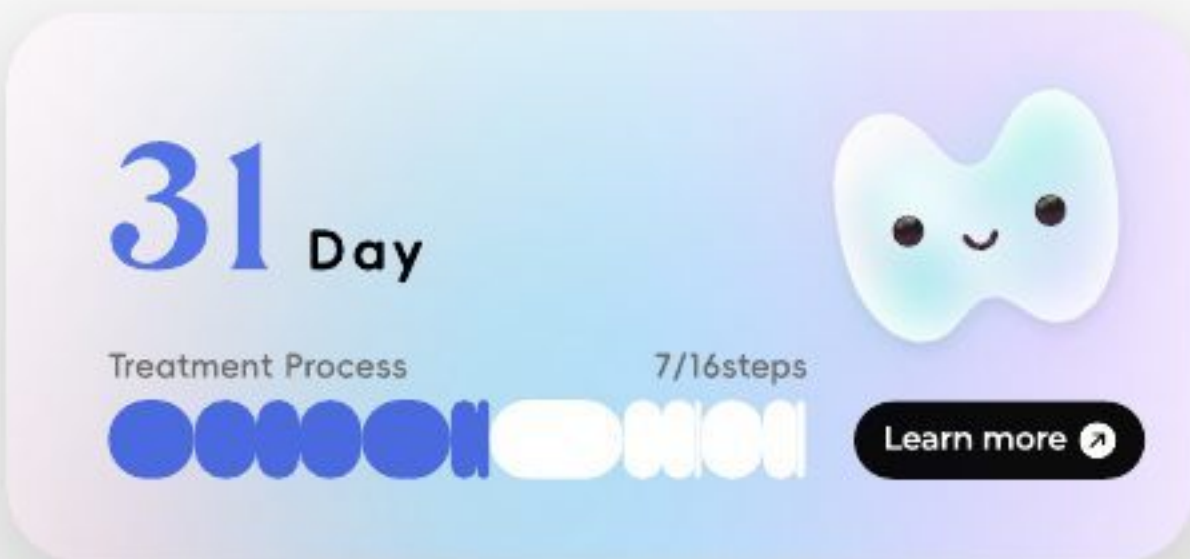
Health Widget

Log record, process preview, chatbot call all in Thyra



Diagnosis Overview

Here, you can get an overview of the entire diagnostic and treatment process.



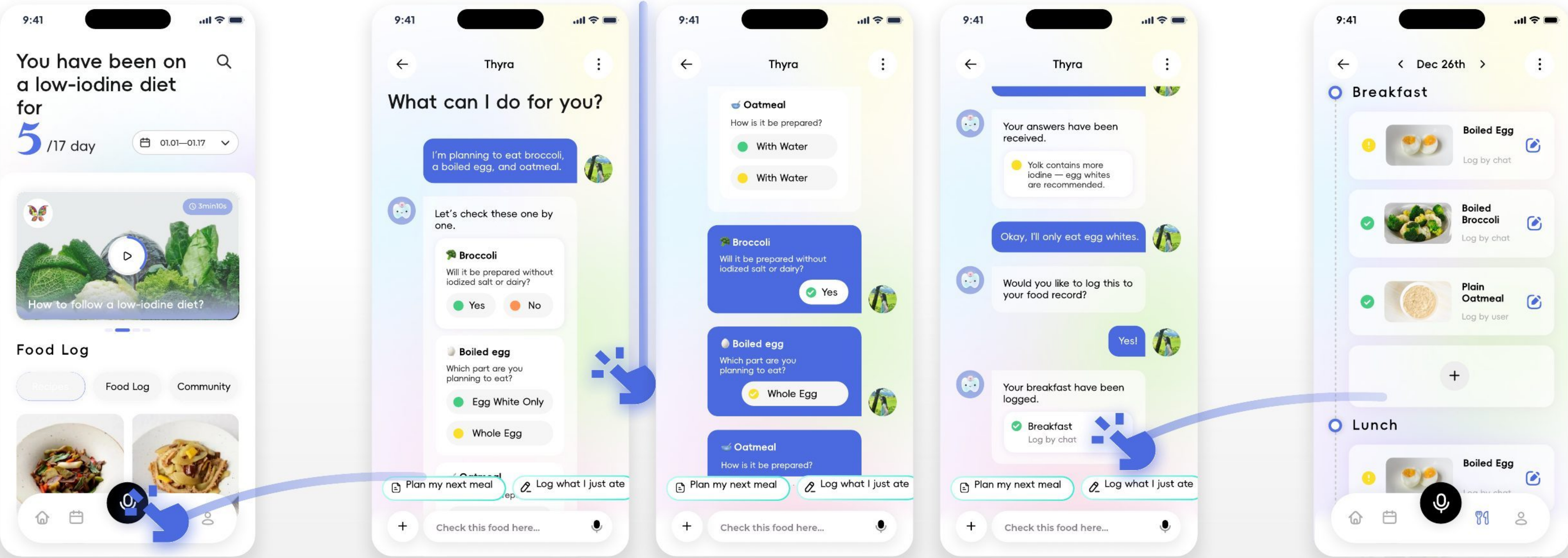
Treatment management

Click on the "Today's Schedule", and you will be able to clearly and thoroughly understand the entire treatment process and tasks for the entire day. Everything has never been so clear and simple.

Schedule Module

Thyra integrates with your phone's native system, making your diagnosis process clearly visible.

Low iodine diet Assistant



Thyra 希芮 Diet Module

Here, you can complete the learning and recording of your diet.

Thyra Diet Chatbot

Not confident about the food you've eaten? Try out the Thyra chatbot! Input the name of the food you ate / Upload the packaging of the food, and Thyra will work with you to complete the low-iodine diet task!

Food Log

Communicate with Thyra to upload your food log, or you can also add it by yourself.